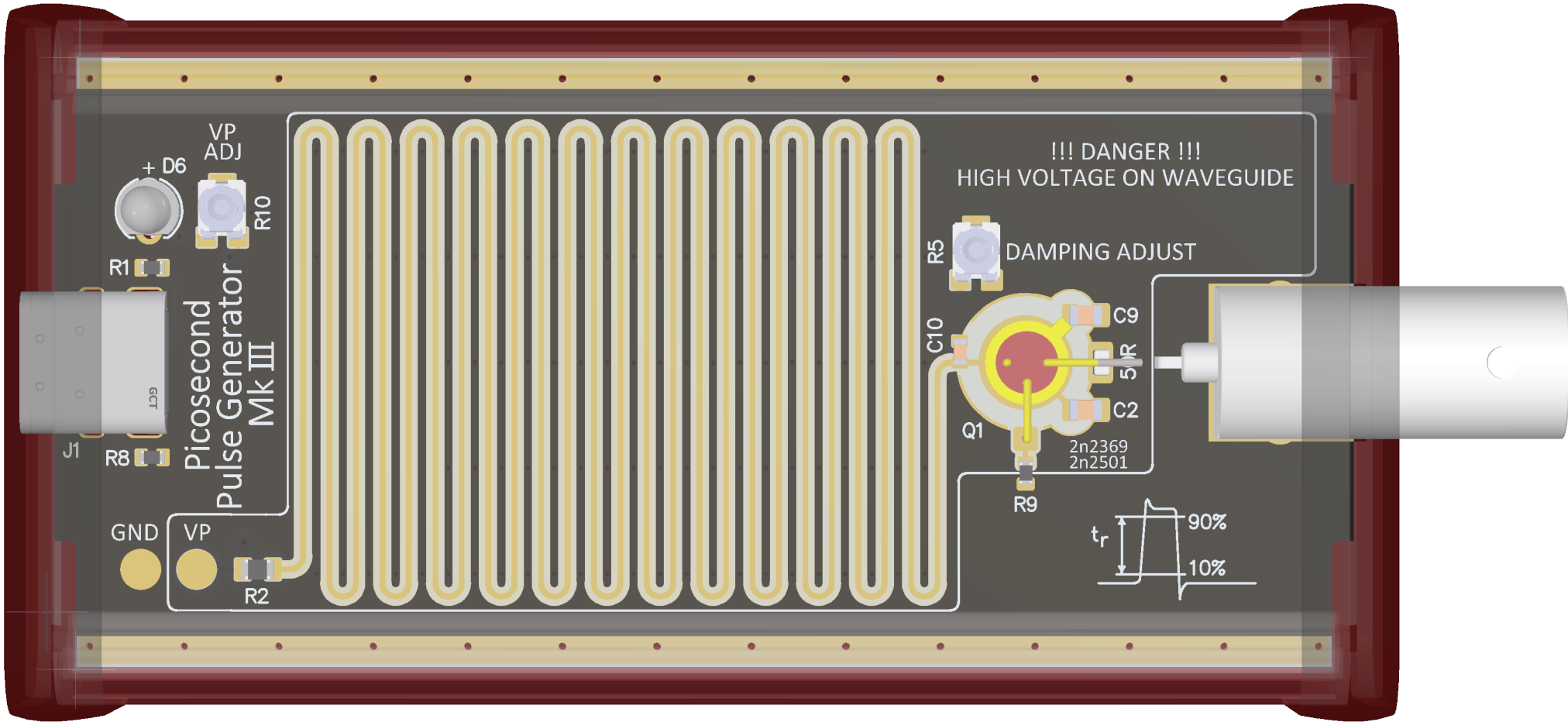


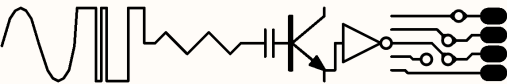
JWpulsegen-V3.PrjPCB



Document Creation Date: 9/25/2025

Design : Vincent Himpe

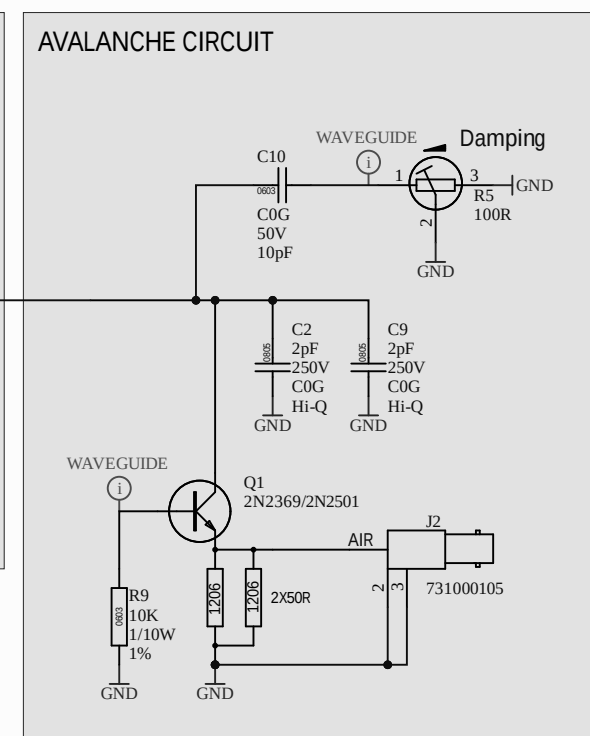
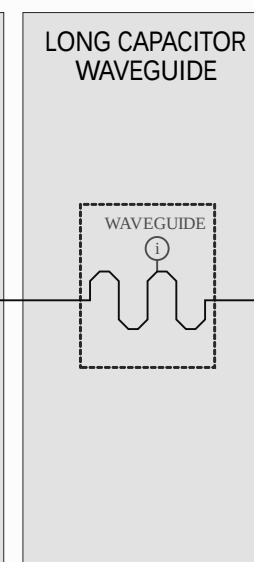
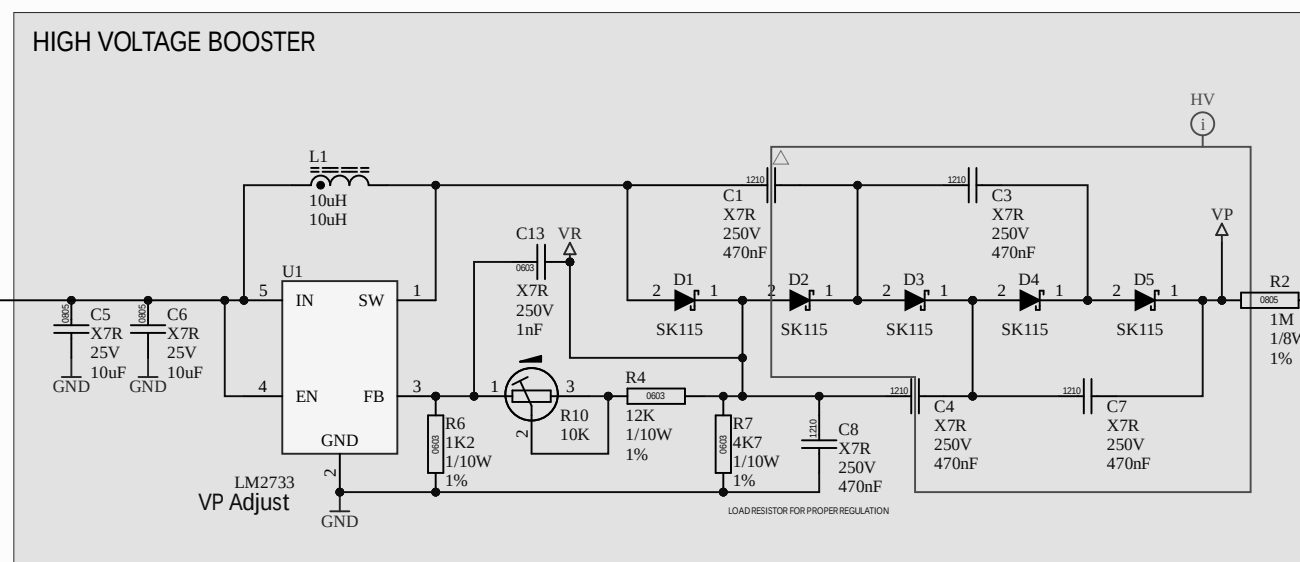
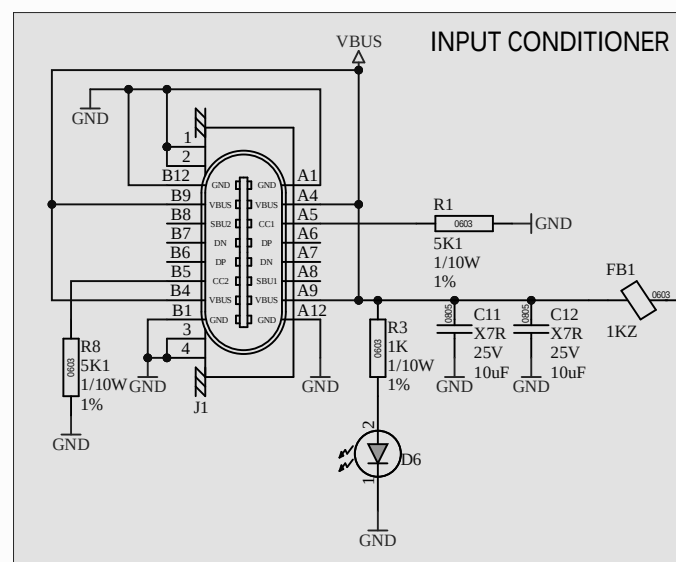
SILICONVALLEYGARAGE



$$VP = VR \times 5$$

$$VR = \frac{1.23V}{1K2} \times (1K2 + R10 + R4)$$

$$C13 : \frac{1}{2 \times 3.14 \times (R4 + R10) \times 8000}$$



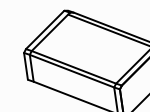
SILICONVALLEYGARAGE

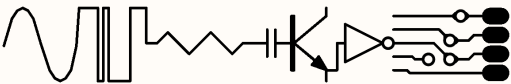
Instruments

PPG3
Picosecond
Pulse Generator

GitHub :
<https://github.com/vincent-himpe/Jim-Williams-PulseGenerator>

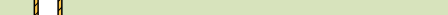
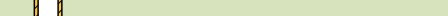
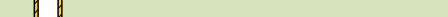
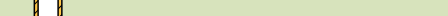
M1
ENCLOSURE-HAMMOND-1455D801

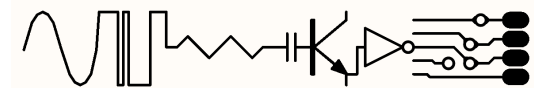


1	2	3	4	5	6
GENERAL					
A	GENERAL 1. DO NOT ALTER SUPPLIED COPPER OR DRILL DATA 2. NO COPPER BALANCING OR REMOVAL OF UNUSED PADS ALLOWED. 3. SILKSCREEN MAY BE CLIPPED / TRIMMED TO EXPOSE COPPER 4. PCB DESIGN AND ACCEPTANCE CRITERIA SHALL FOLLOW THE REQUIREMENTS OF IPC-2221, IPC-2222, AND IPC-6012 CLASS 2 5. ALL SPECIFICATIONS SHALL BE THE LATEST STANDARDS, UNLESS OTHERWISE NOTED 6. ALL MODIFICATIONS MUST BE COMMUNICATED AND APPROVED IN WRITING.				
	MATERIALS 7. MATERIALS SHALL BE ACCORDING TO THE STACKUP DRAWING IN THIS DOCUMENT. 8. MATERIAL SHALL HAVE A FLAMABILITY RATING OF UL 94V-0 OR BETTER 9. SURFACE FINISH : HASL 10. SOLDER MASK COLOR : BLACK 11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm 12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL 13. THICKNESS LISTED IN LAYER STACK LEGEND REPRESENT FINAL PRESSED VALUES FOR THE PREPREG 14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
D	QA, ELECTRICAL TEST AND MARKINGS 15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
SILICONVALLEYGARAGE 				Project JWpulsegen-V3.PrjPCB Version: Variant [No Variations]	
				FABRICATION DRAWING	
1	2	3	4	5	6

LAYER STACK

Layer Stack Legend

	Material	Layer	Thickness	Dielectric Material	Type	Gerber	Dk	Weight	Constructions	Df	Resin
		Top Overlay			Legend	GTO					
	Surface Material	Top Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GTS	3.5				
	Copper	Top Layer	0.036mm(1.400mil)		Signal	GTL		1oz			
	Prepreg		0.210mm(8.284mil)	7628 X 1	Dielectric		4.5		1080	0.02	62%
	CF-004	Layer 1	0.015mm(0.598mil)		Signal	G1		0.5oz			
	Core		1.065mm(41.929mil)	FR-4	Dielectric		4.8				
	CF-004	Layer 2	0.015mm(0.598mil)		Signal	G2		0.5oz			
	Prepreg		0.210mm(8.284mil)	7628 X 1	Dielectric		4.5		1080	0.02	62%
	Copper	Bottom Layer	0.036mm(1.400mil)		Signal	GBL		1oz			
	Surface Material	Bottom Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GBS	3.5				
		Bottom Overlay			Legend	GBO					
Total thickness: 1.608mm(63.293mil)											



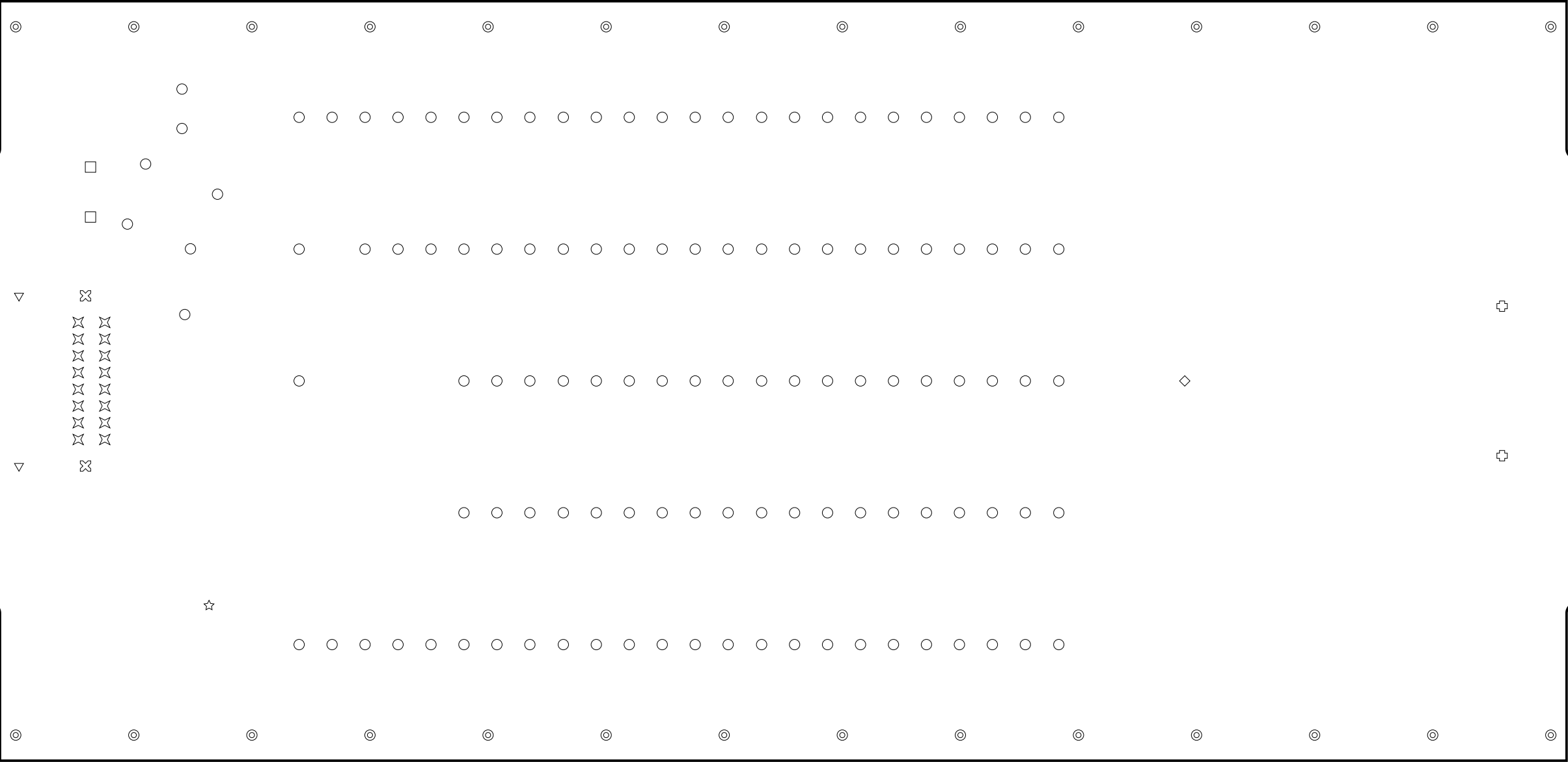
DRILL LEGEND

Drill Table

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Via / Pad	Pad Shape	Description	Hole Tolerance	Via Type	Via Feature
○	117	0.300mm(11.811mil)	Plated	Round	Top Layer - Bottom Layer	Via				Type 1B	Tenting Both
☆	1	0.356mm(14.000mil)	Plated	Round	Top Layer - Bottom Layer	Via				Type 1B	Tenting Both
✕	16	0.400mm(15.748mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
◎	28	0.500mm(19.685mil)	Plated	Round	Top Layer - Bottom Layer	Via				Type 1B	Tenting Both
□	2	1.000mm(39.370mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
▽	2	1.400mm(55.118mil)	Plated	Slot	Top Layer - Bottom Layer	Pad	Rounded Rectangular				
⊕	2	1.800mm(70.866mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
⊗	2	2.100mm(82.677mil)	Plated	Slot	Top Layer - Bottom Layer	Pad	Rounded Rectangular				
◇	1	5.000mm(196.850mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	171 Total										

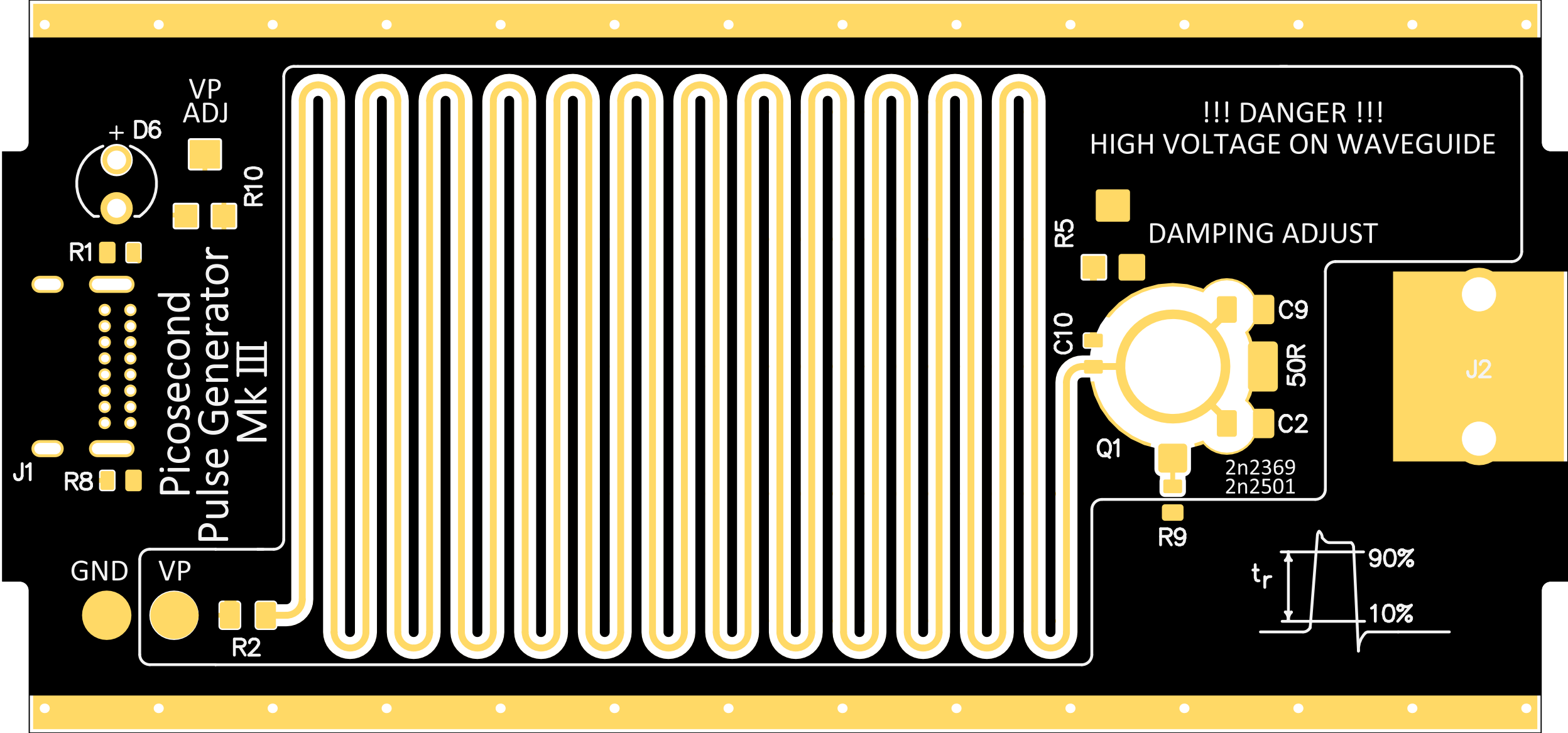
DRILL DRAWING

Drill Drawing View (Scale 4:1)



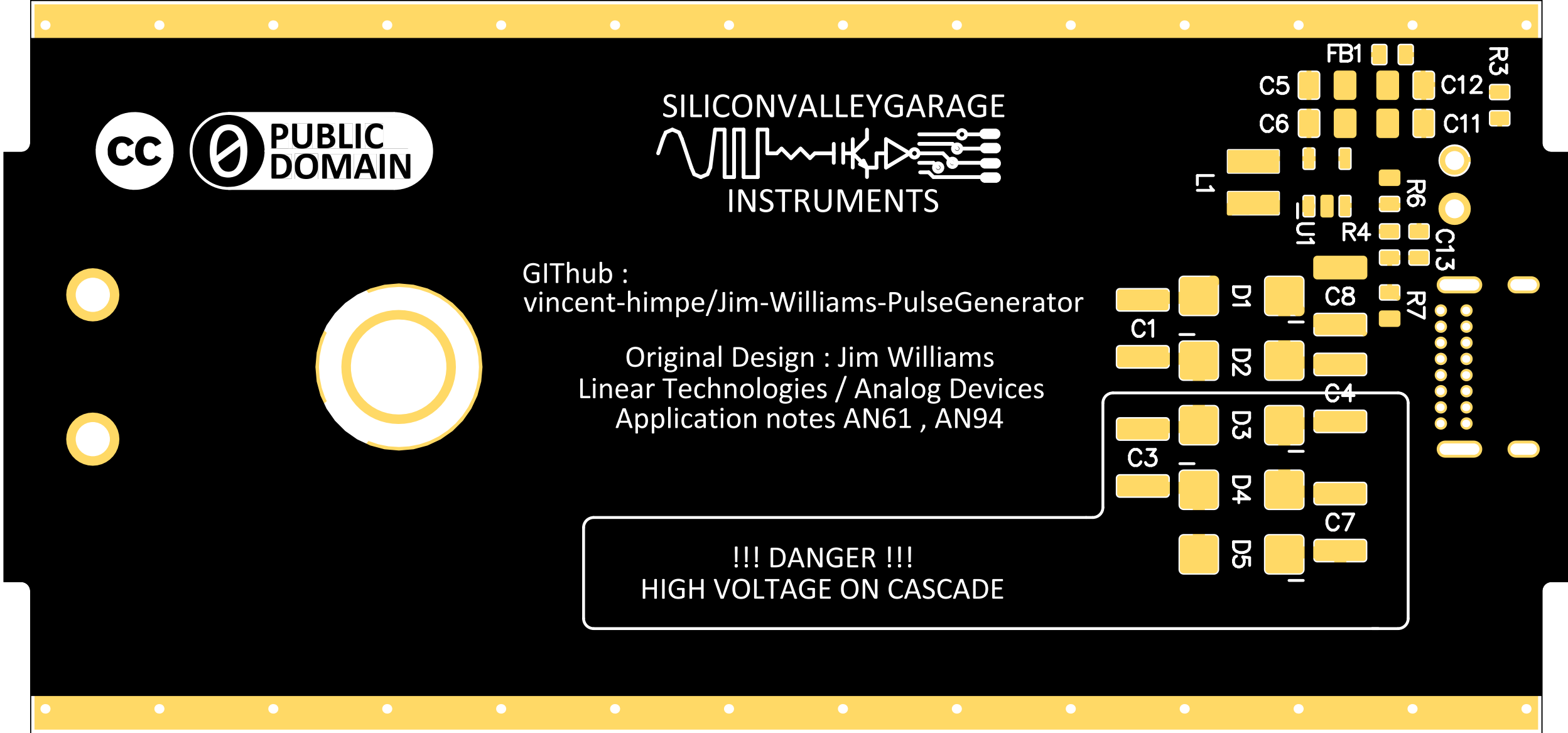
COMPOSITE VIEW FRONT

Top Layer (Scale 4:1)



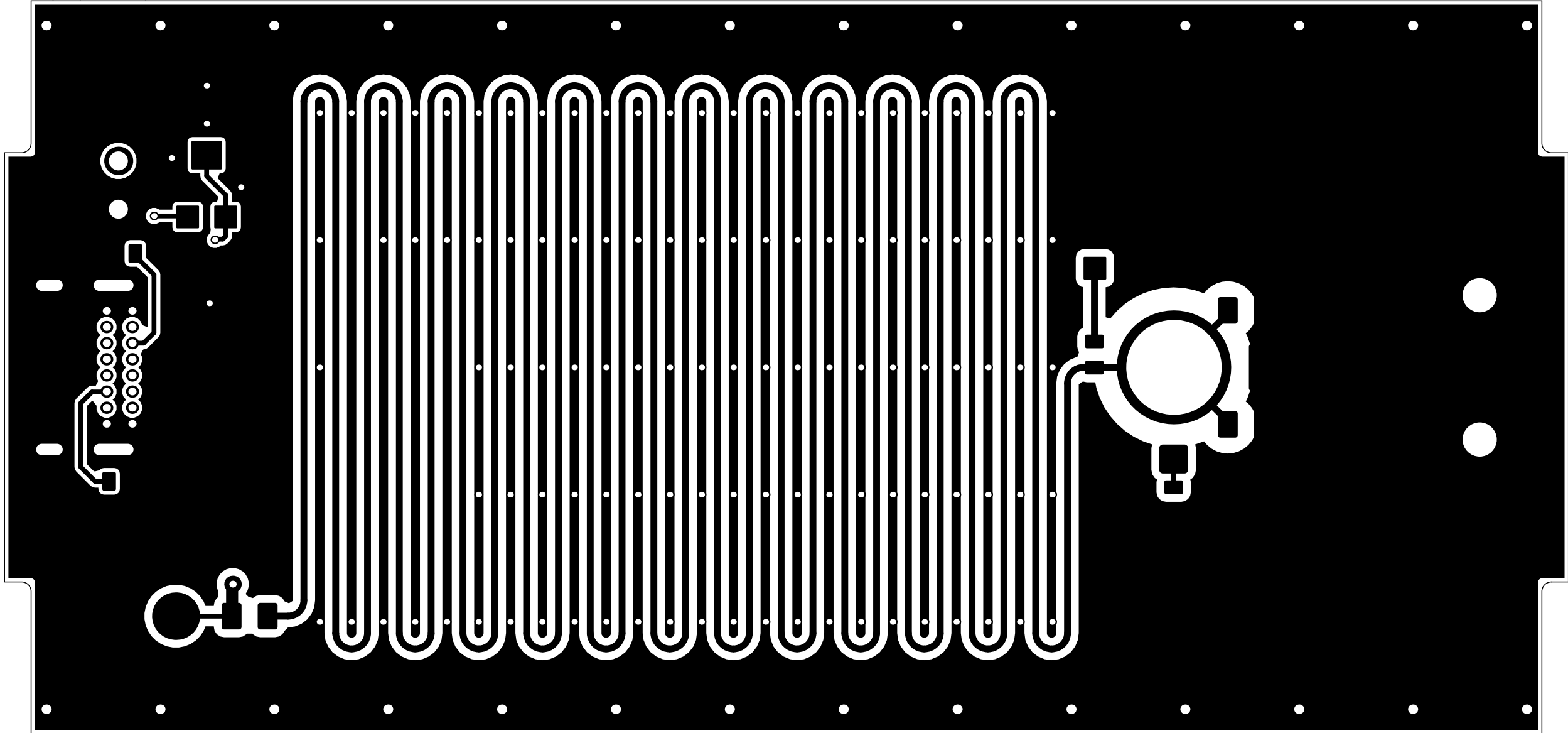
COMPOSITE VIEW BACK

Bottom Layer (Scale 4:1)



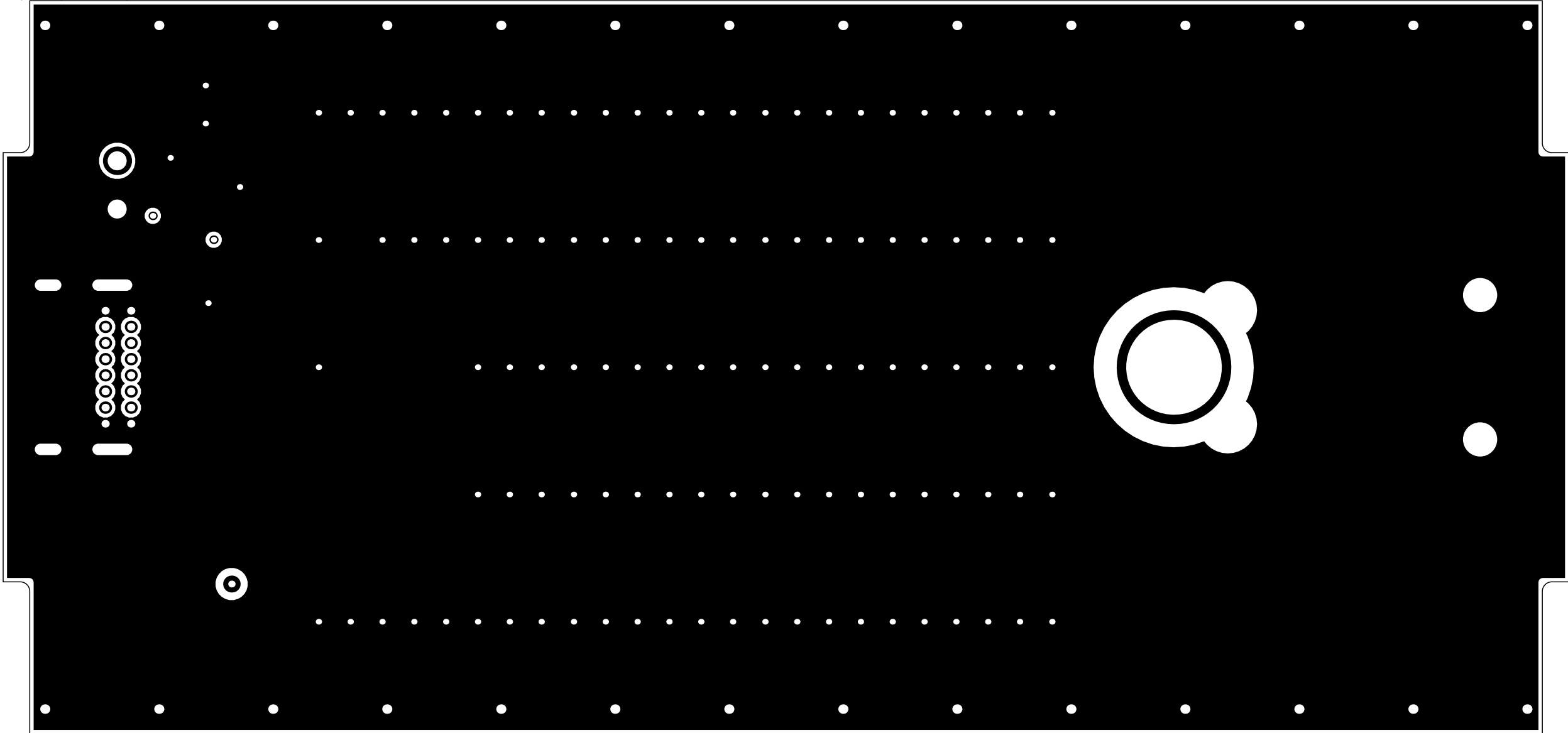
LAYER VIEW : TOP LAYER

Top Layer (Scale 4:1)

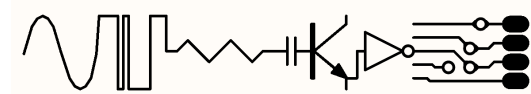


LAYER VIEW : MID LAYER 1

Layer 1 (Scale 4:1)



SILICONVALLEYGARAGE



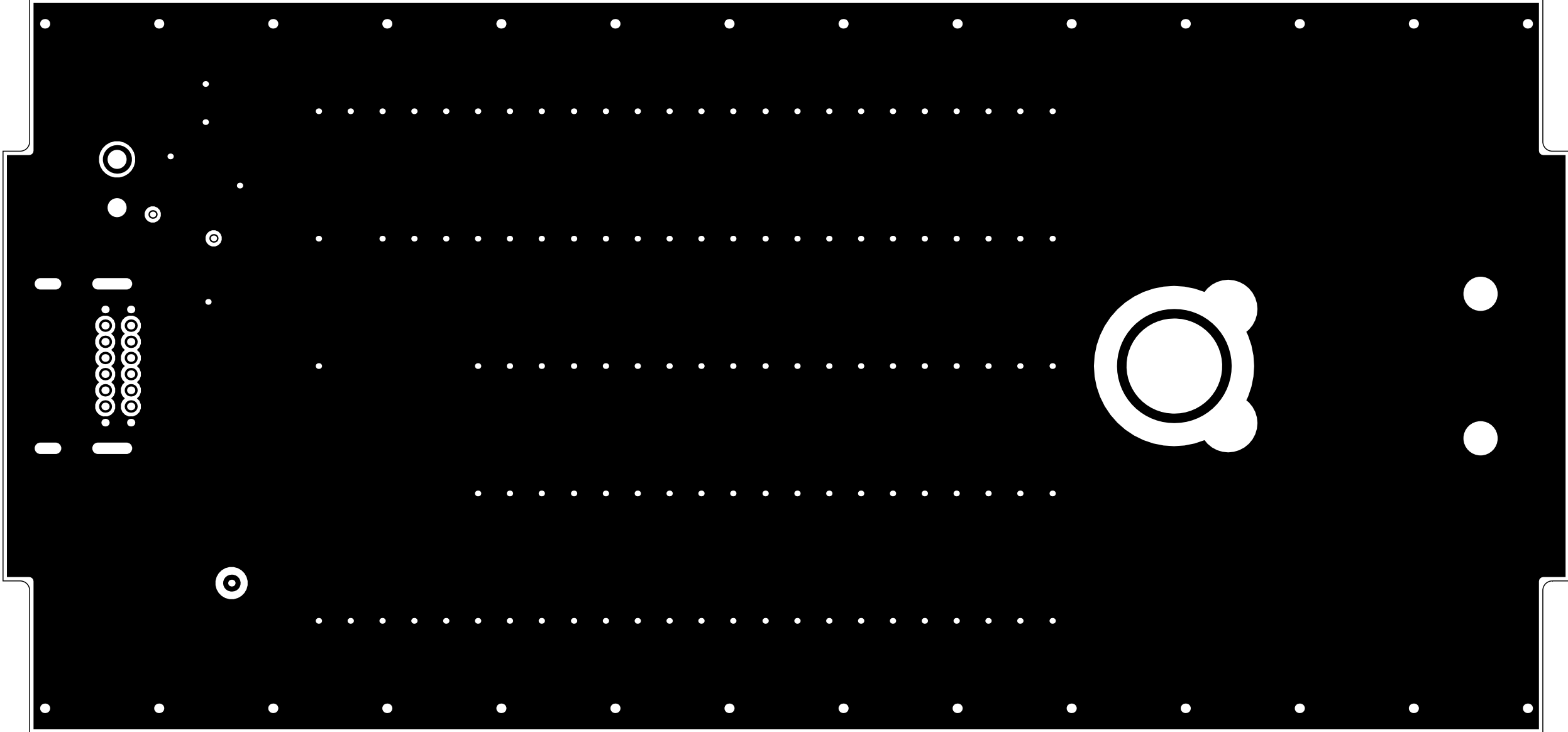
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

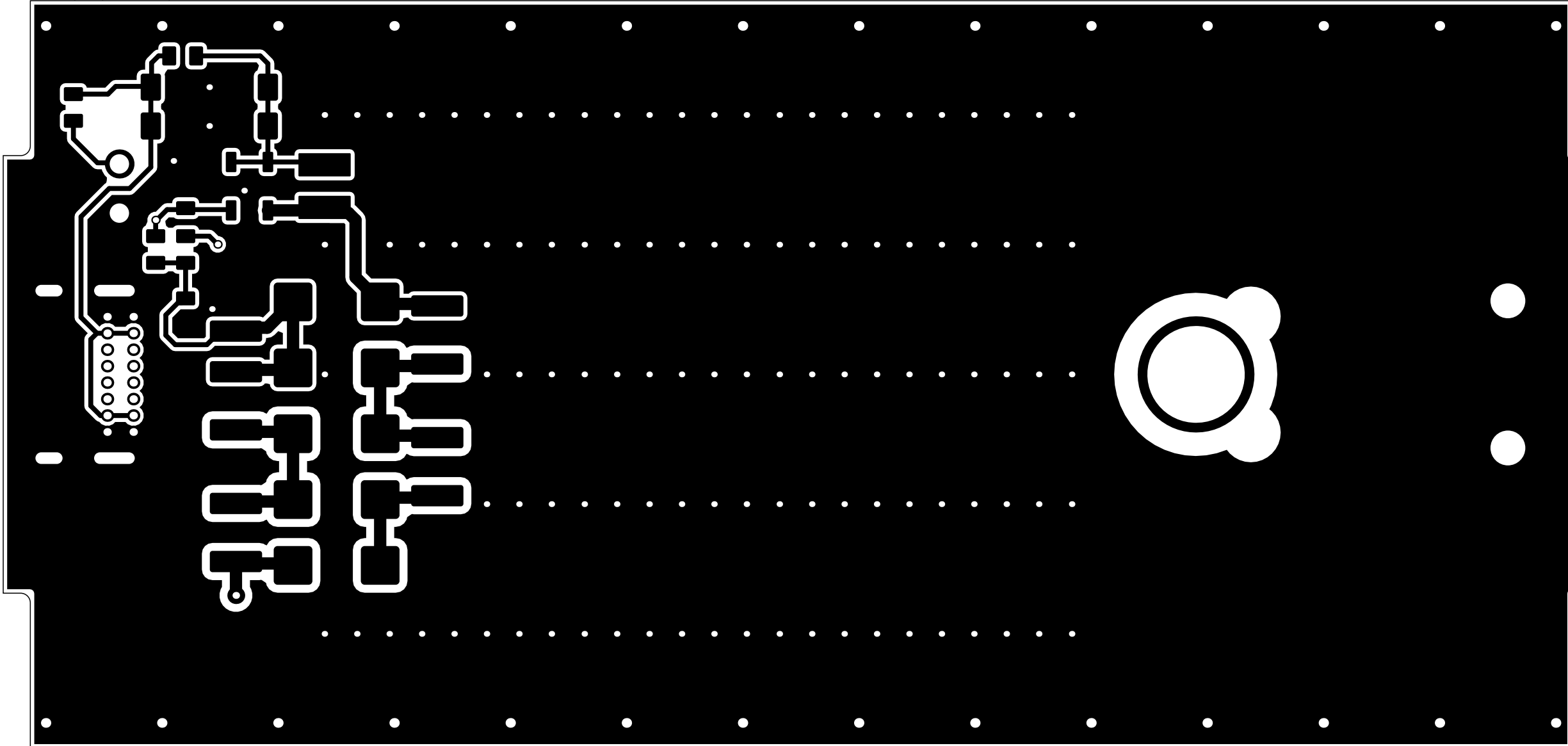
LAYER VIEW : MID LAYER 2

Layer 2 (Scale 4:1)



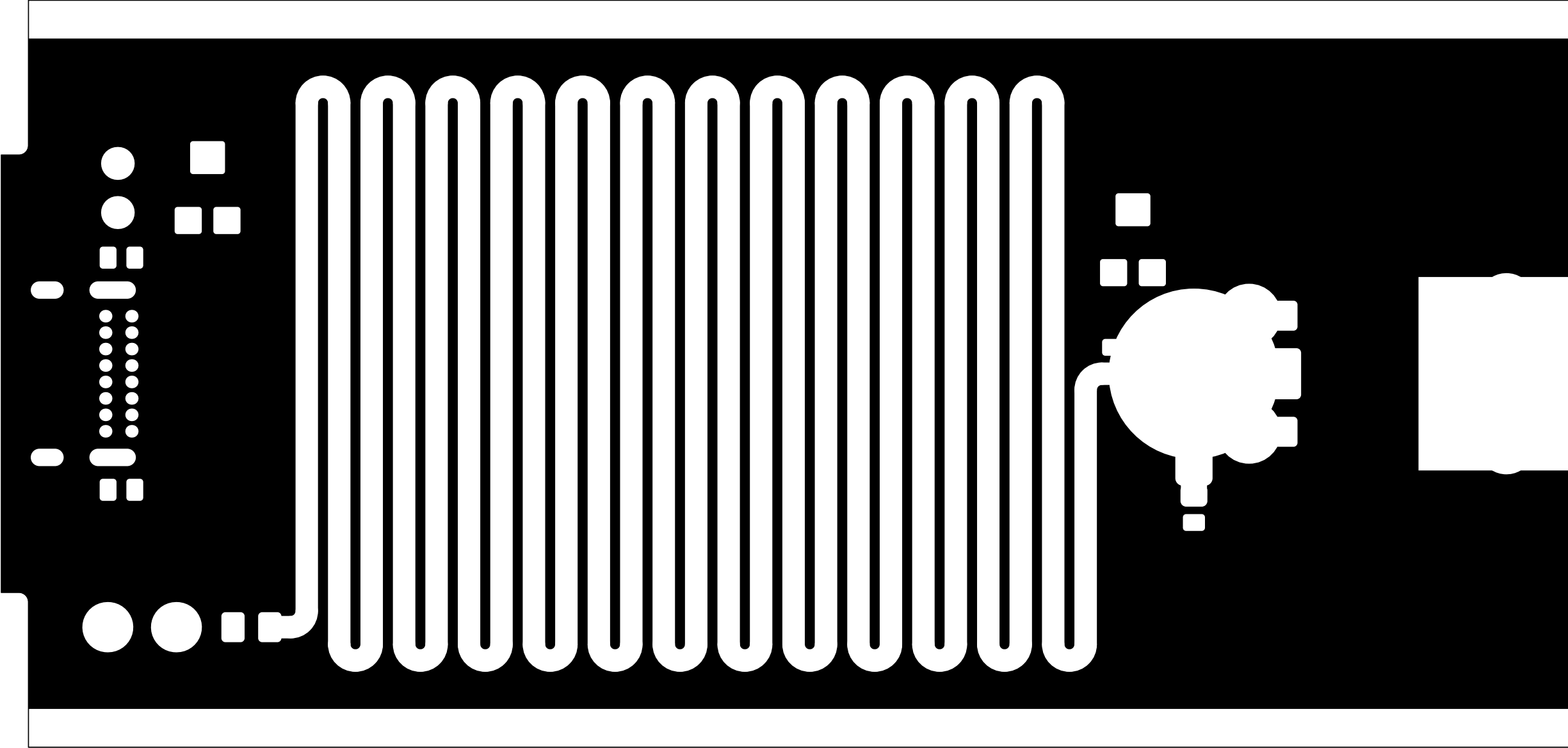
LAYER VIEW : BOTTOM LAYER

Bottom Layer (Scale 4:1)



LAYER VIEW : TOP SOLDER MASK

Top Solder (Scale 4:1)



73

74

75

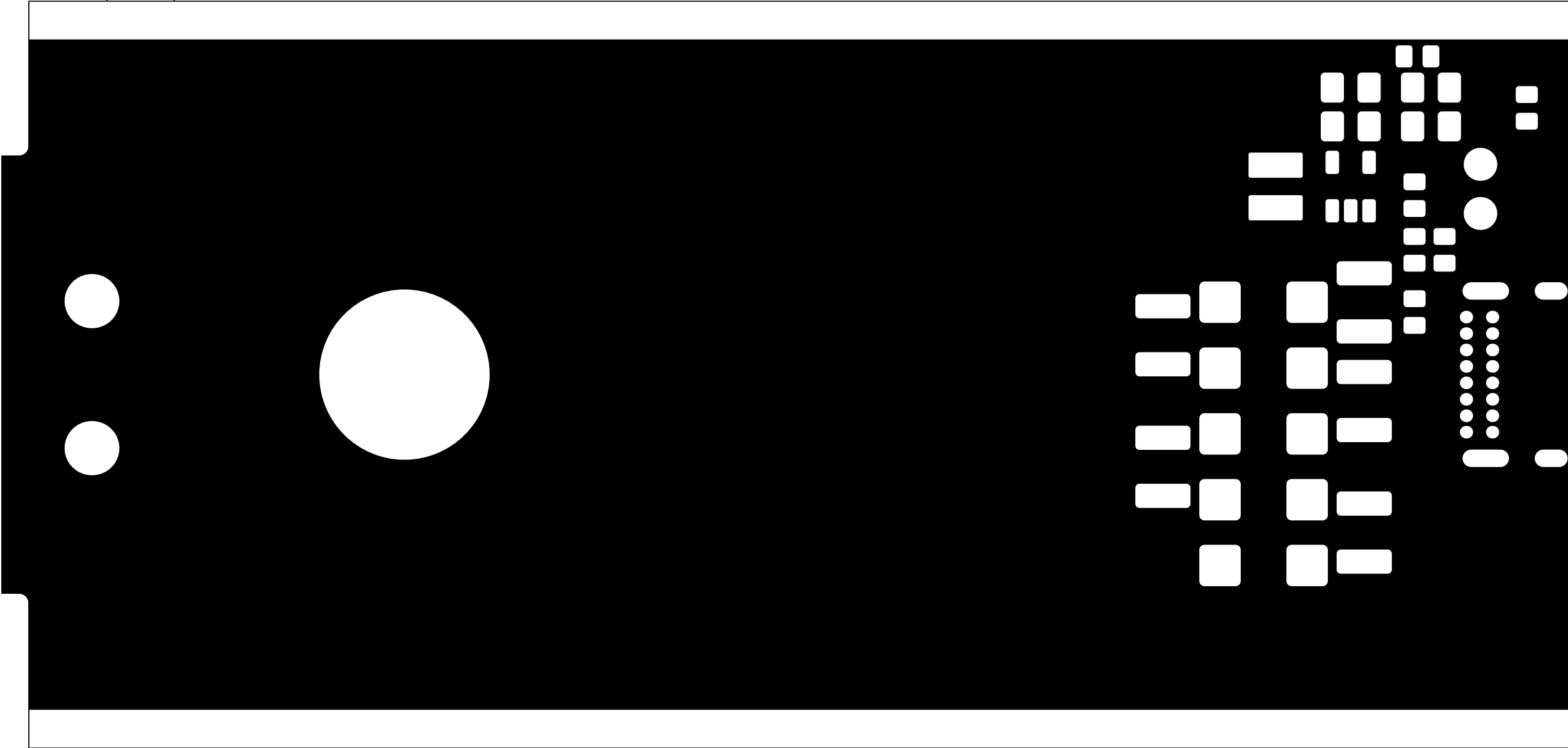
76

77

78

LAYER VIEW : BOTTOM SOLDER MASK

Bottom Solder (Scale 4:1)



73

74

75

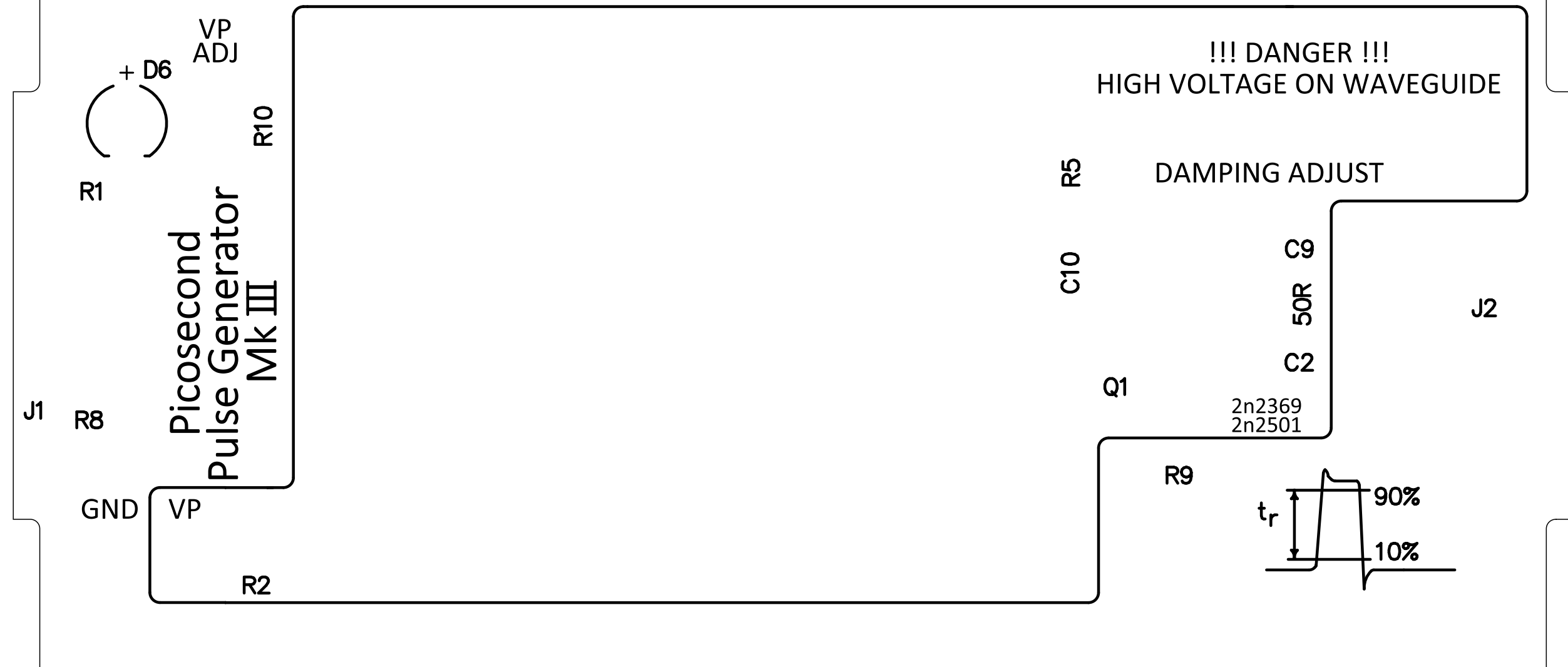
76

77

78

LAYER VIEW : TOP SILKSCREEN (LEGEND)

Top Overlay (Scale 4:1)

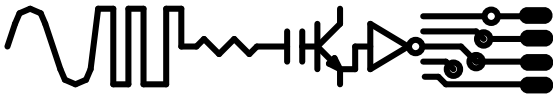


LAYER VIEW : BOTTOM SILKSCREEN (LEGEND)

Bottom Overlay (Scale 4:1)



SILICONVALLEYGARAGE
INSTRUMENTS



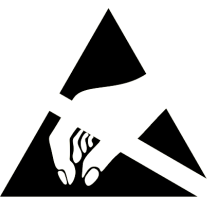
GIThub :
vincent-himpe/Jim-Williams-PulseGenerator

Original Design : Jim Williams
Linear Technologies / Analog Devices
Application notes AN61 , AN94

!!! DANGER !!!
HIGH VOLTAGE ON CASCADE

FB1
C5
C6
C12
C11
R3
C13
R6
R7
R4
C8
C4
C7
D1
D2
D3
D4
D5
C1
C3
L1
U1

GENERAL



Unless otherwise specified the following rules apply:

- 1. DO NOT DEVIATE FROM ARTWORK OR BOM WITHOUT PRIOR AUTHORIZATION.
- 2. ASSEMBLE AND INSPECT PER IPC-610 CLASS 2

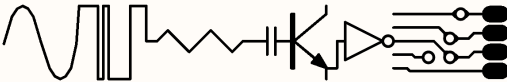
Bill of Materials and Material Handling

- 3. THE BOM CONTAINED IN THIS DOCUMENT IS AS-BUILT. NON-INSTALLED PARTS HAVE BEEN REMOVED. ADDITIONAL BOM FORMATS ARE AVAILABLE IN THE PROJECT FILES
- 4. ANY PART SUBSTITUTIONS MUST BE APPROVED IN WRITING BEFORE ASSEMBLY
- 5. ALL MATERIALS MUST BE PROCURED FROM MANUFACTURER AUTHORIZED DISTRIBUTORS OR THE ORIGINAL MANUFACTURER
- 6. ALL COMPONENTS AND BOARDS TO BE HANDLED AND STORED ACCORDING TO IPC GUIDELINES
- 7. ESD CONTROL PER IPC RULES

Soldering

- 8. SOLDERING TO BE DONE USING SN37PB63 ALLOY USING ALLOY MANUFACTURER RECOMMENDED NO-CLEAN FLUX
- 9. BGA COMPONENTS WITH LEAD-FREE CONNECTIONS NEED TO BE REBALLED WITH SN63PB37. MIXING OF ALLOYS IS NOT PERMITTED.
- 10. SOLDERING PREFERRABLY TO BE DONE USING NITROGEN ATMOSPHERE
- 11. SURPLUS COMPONENTS TO VACUUM SEALED WITH DESSICANT IN ANTISTATIC BAGS
- 12. INCOMING MATERIAL (BOARDS AND COMPONENTS) NEEDS TO BE INSPECTED FOR HUMIDITY AND BAKED IF NEEDED PRIOR TO USE.
- 13. MANUAL REWORK / TOUCHUP TO BE DONE USING SAME ALLOY AND APPROPRIATE FLUX. FLUX MUST BE REMOVED.

SILICONVALLEYGARAGE



Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

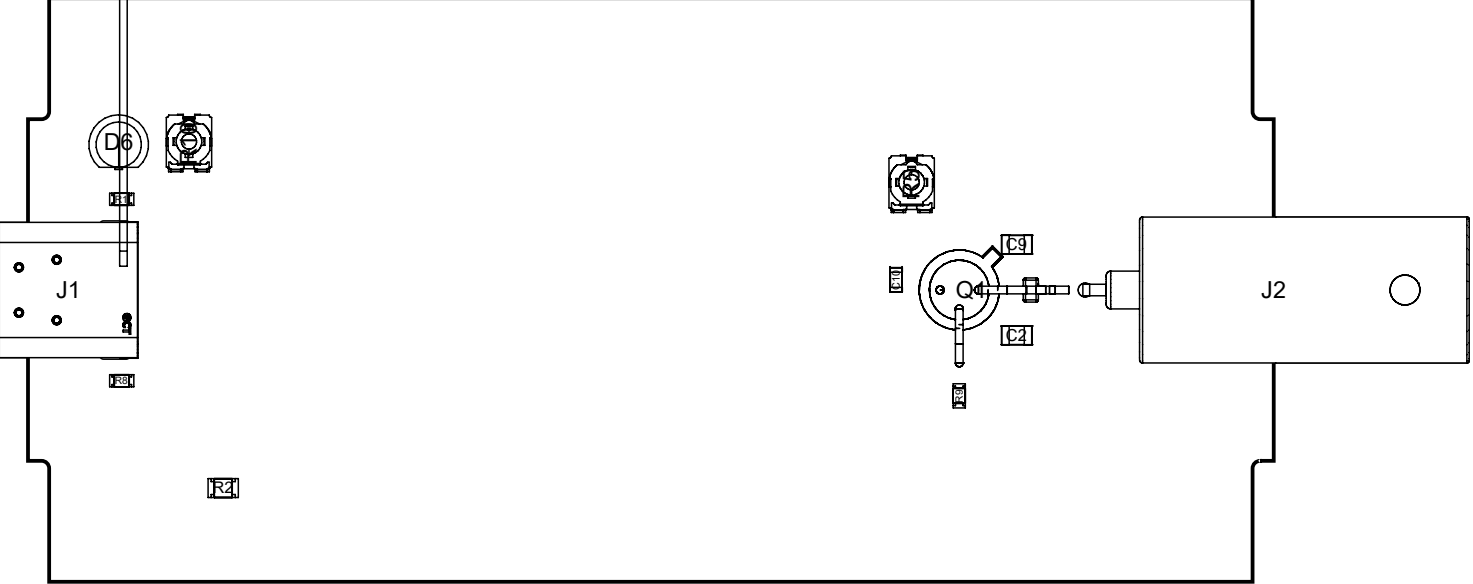
ASSEMBLY DRAWING

2D VIEW

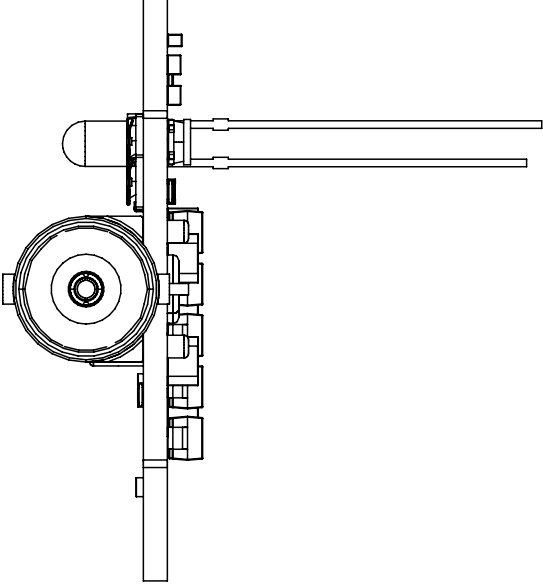
View from Front side (Scale 2:1)



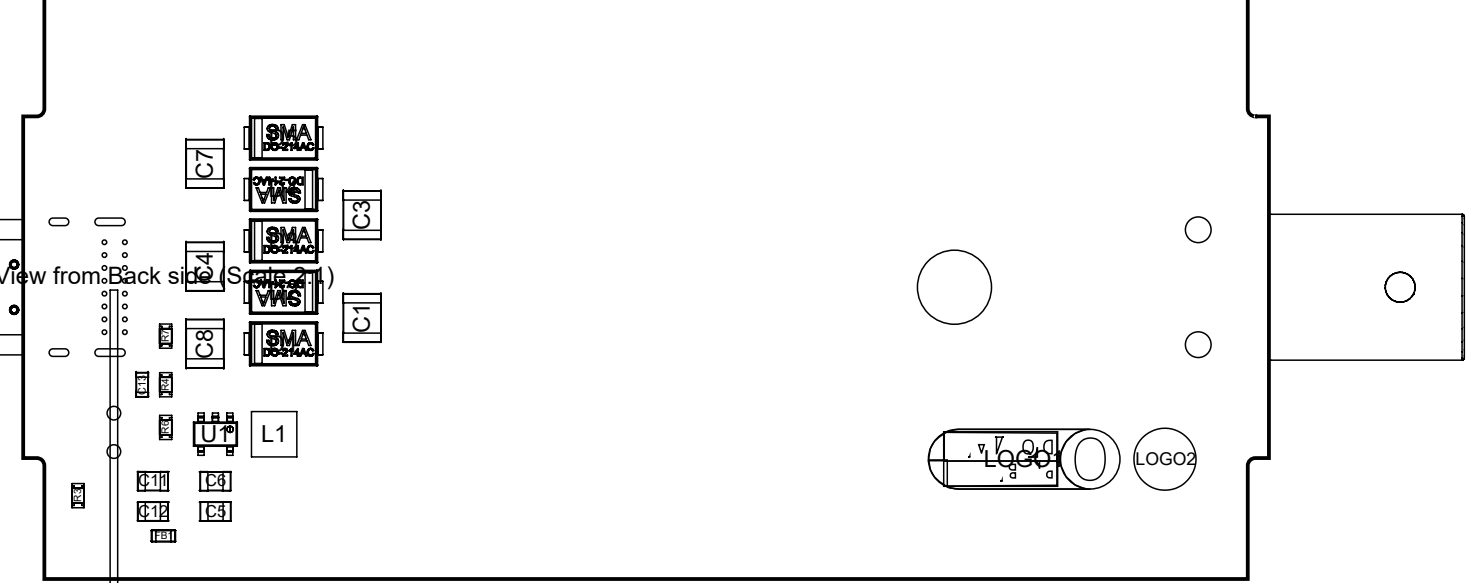
View from Top side (Scale 2:1)



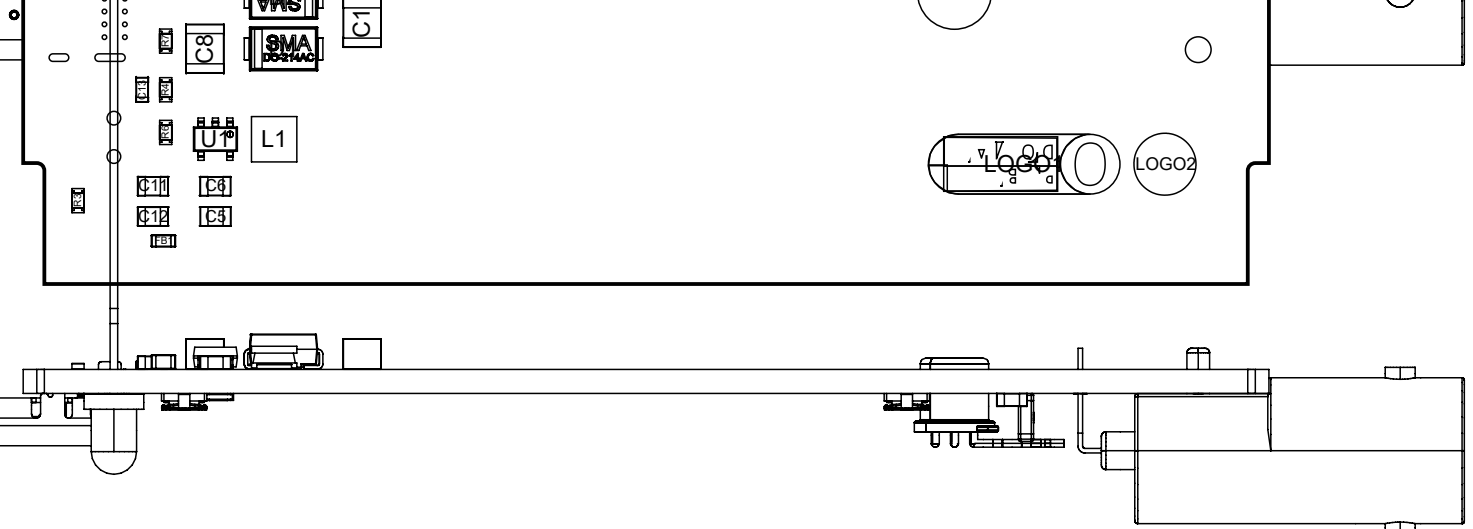
View from Right side (Scale 2:1)



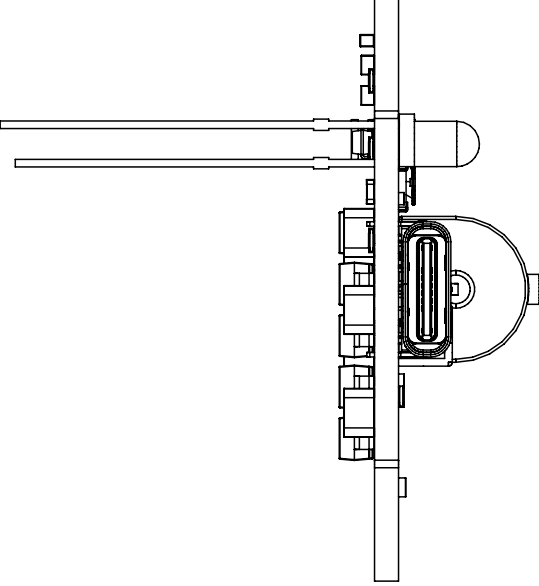
View from Bottom side (Scale 2:1)



View from Back side (Scale 2:1)



View from Left side (Scale 2:1)



SILICONVALLEYGARAGE

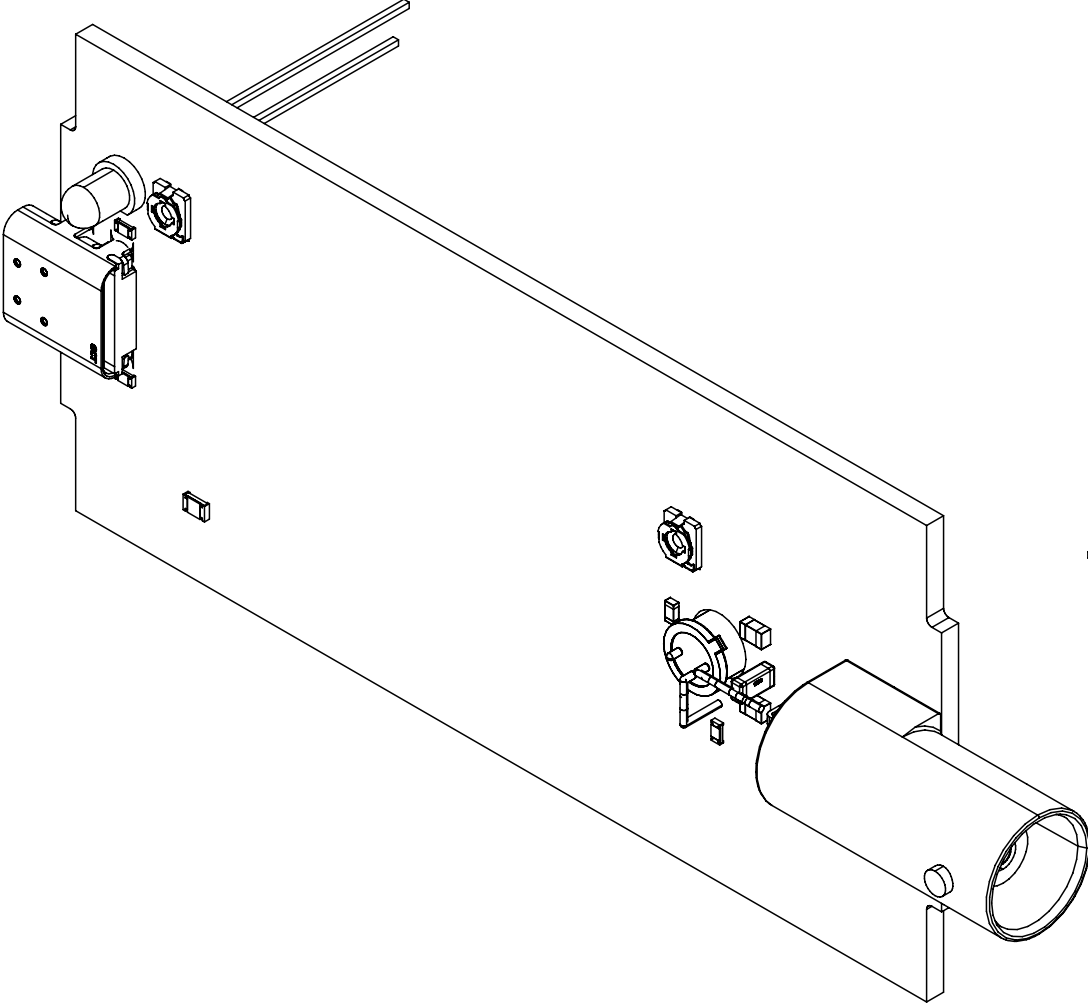
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

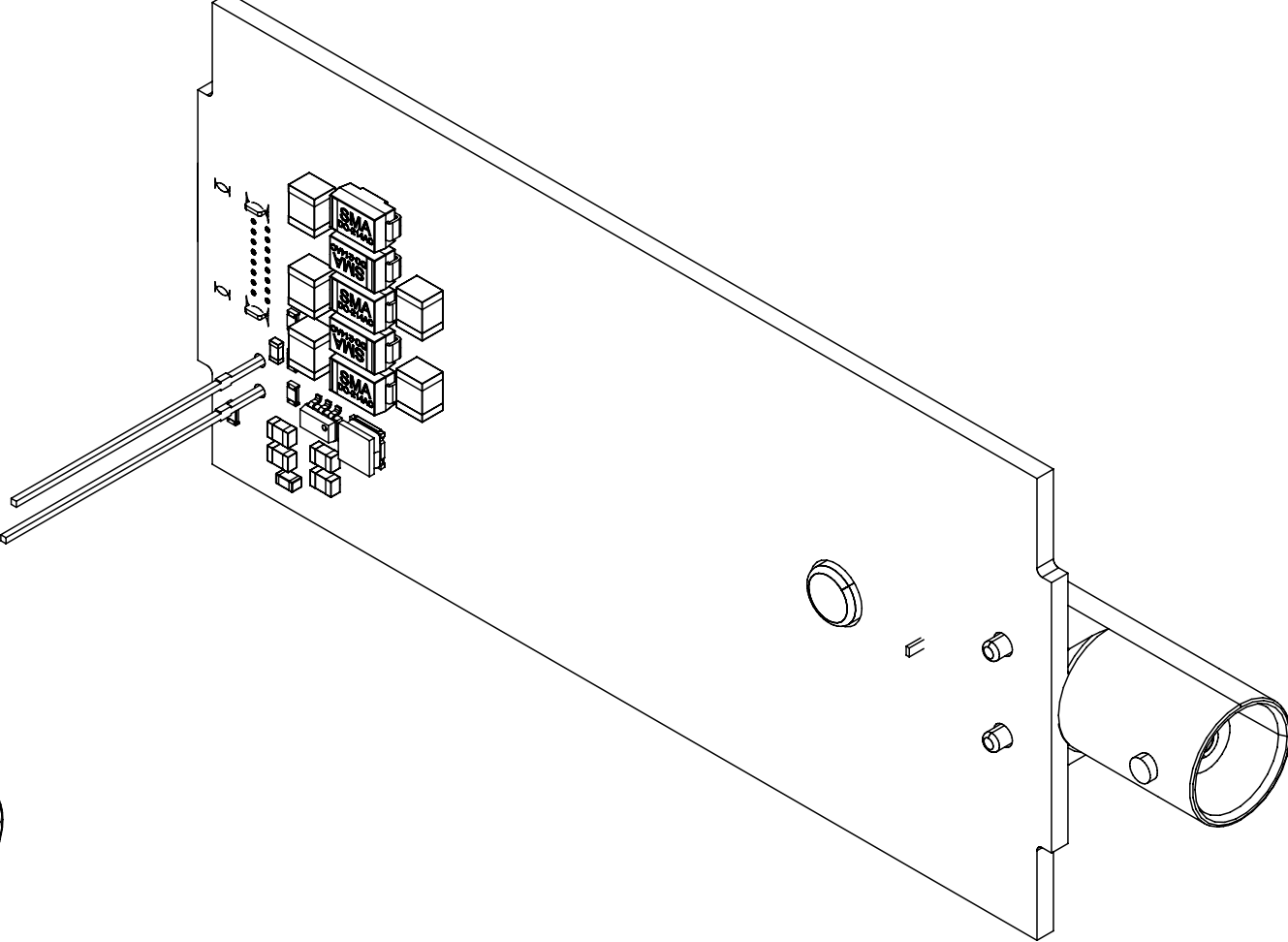
ASSEMBLY DRAWING

3D VIEW

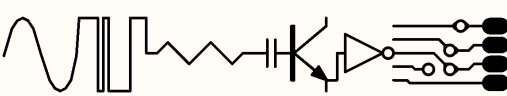
View from Top side (Scale 2:1)



View from Bottom side (Scale 2:1)



SILICONVALLEYGARAGE



Project JWpulsegen-V3.PrjPCB

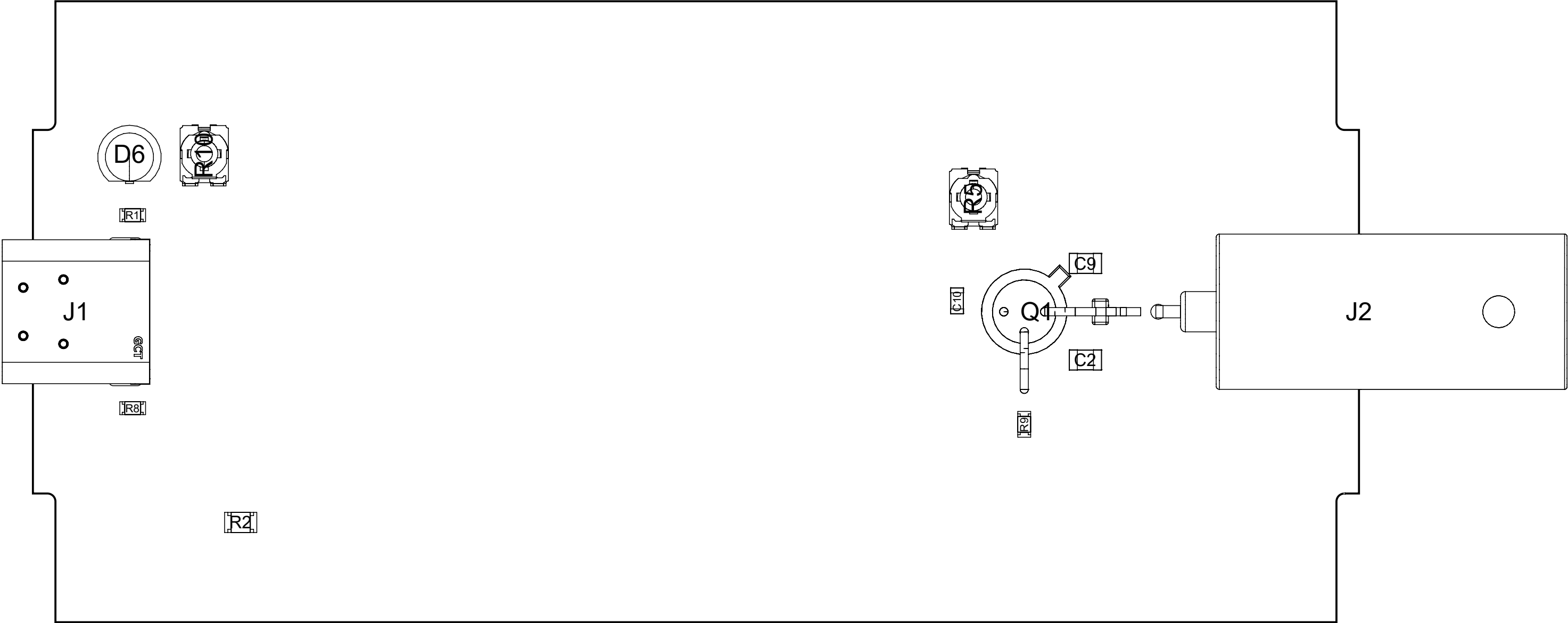
Version: | Variant [No Variations]

ASSEMBLY DRAWING

1	2	3	4	5	6
Bill Of Materials					

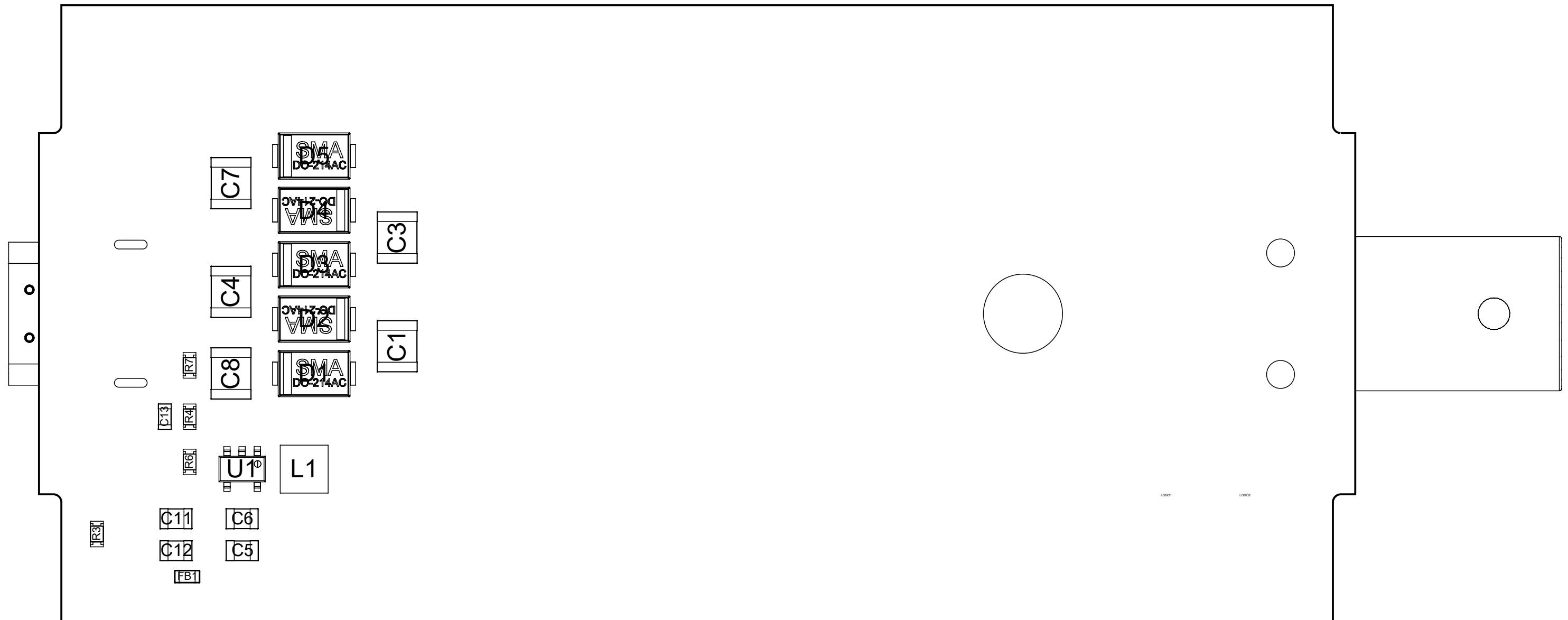
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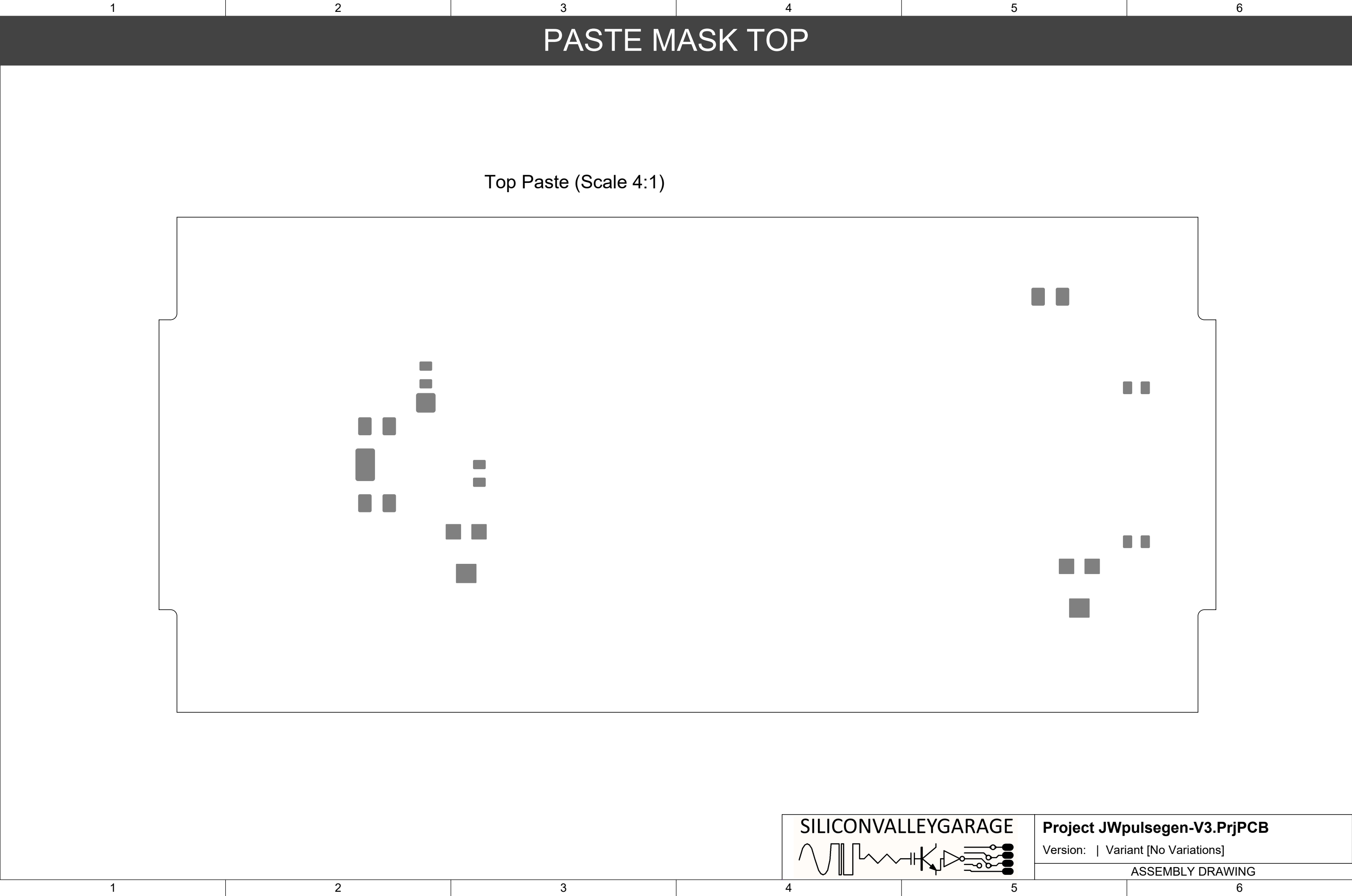
DESIGNATORS FRONT

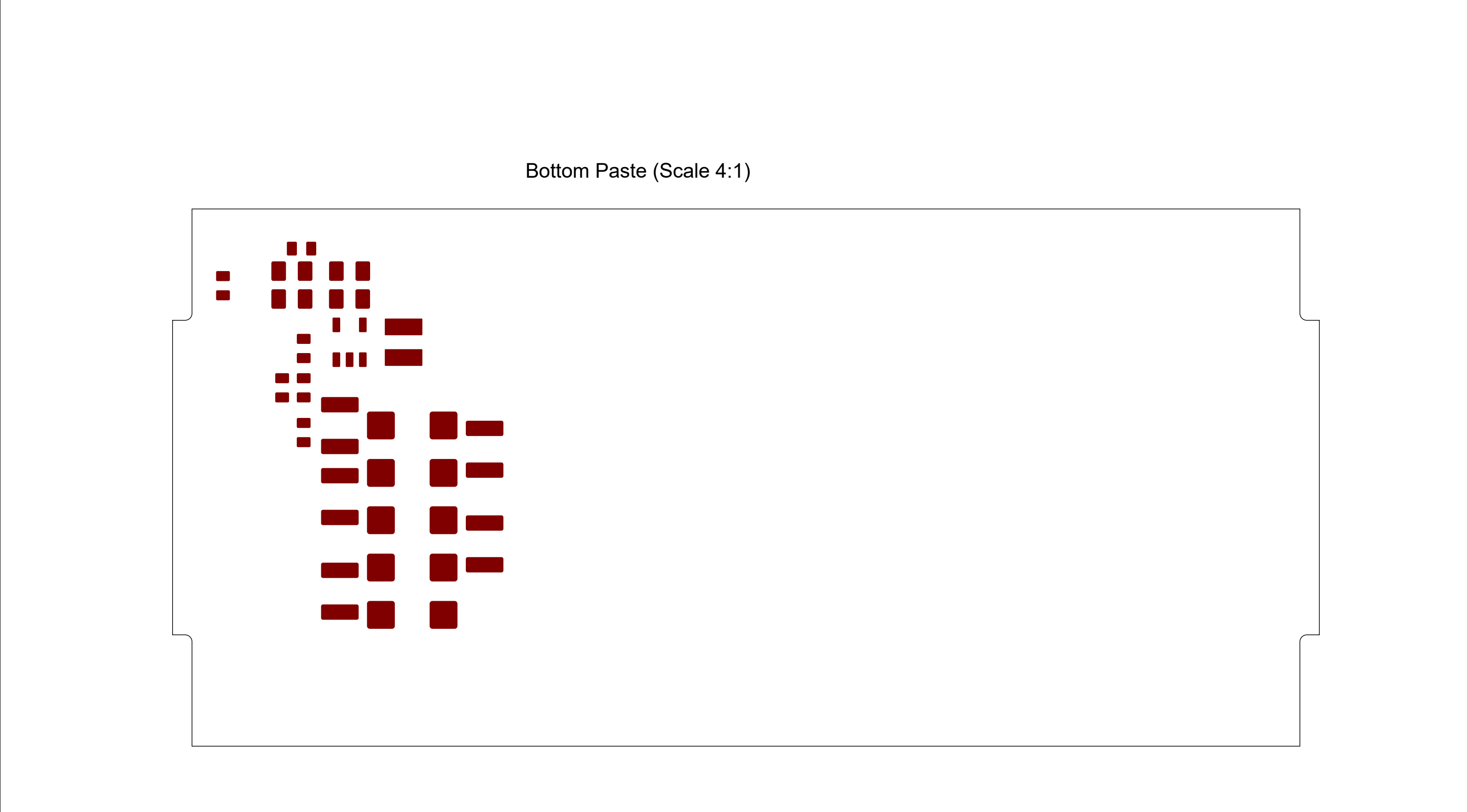


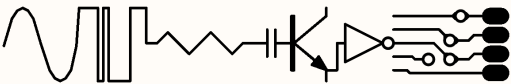
View from Top side (Scale 4:1)

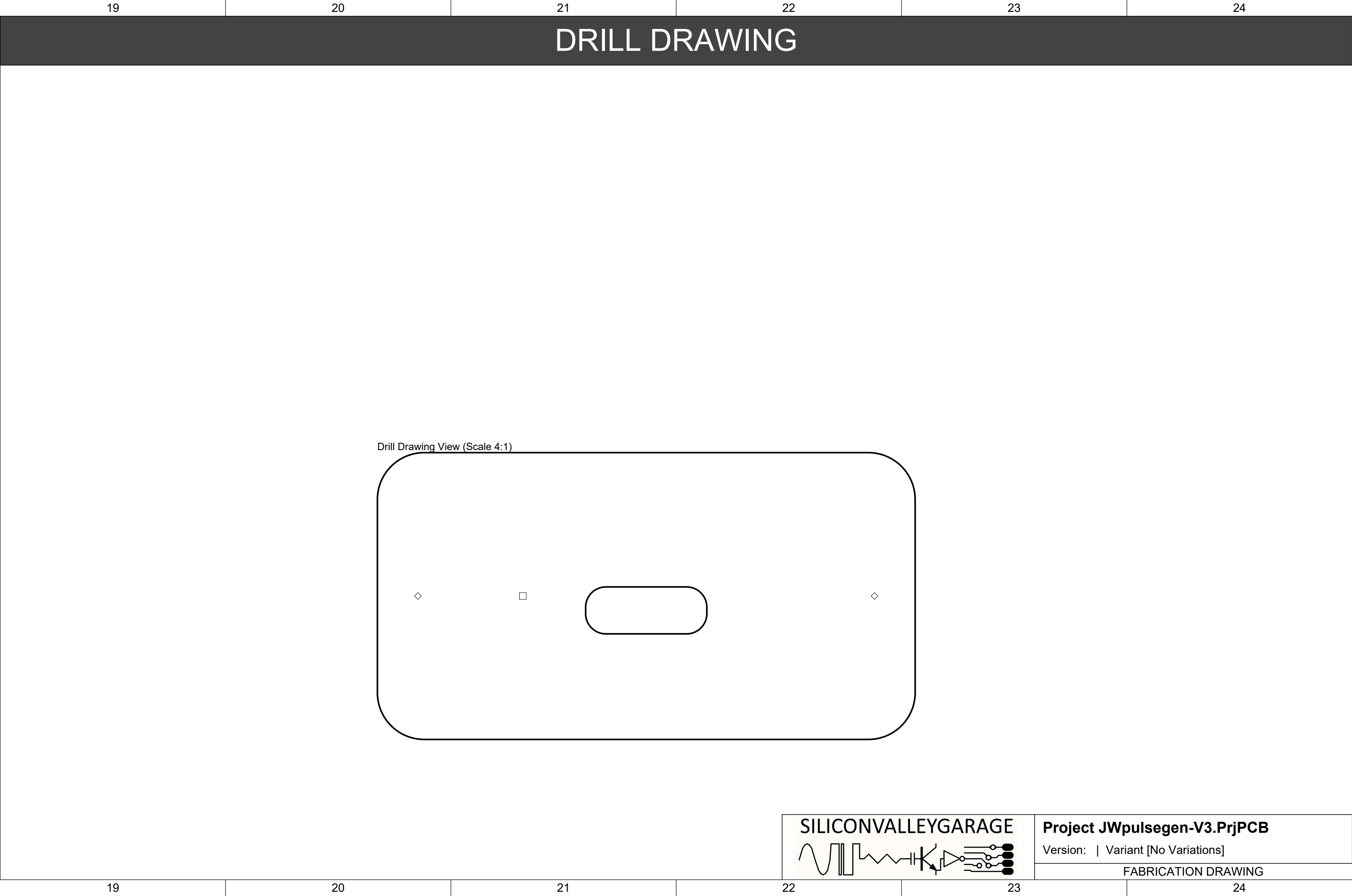
DESIGNATORS BACK







1	2	3	4	5	6
GENERAL					
A	GENERAL 1. DO NOT ALTER SUPPLIED COPPER OR DRILL DATA 2. NO COPPER BALANCING OR REMOVAL OF UNUSED PADS ALLOWED. 3. SILKSCREEN MAY BE CLIPPED / TRIMMED TO EXPOSE COPPER 4. PCB DESIGN AND ACCEPTANCE CRITERIA SHALL FOLLOW THE REQUIREMENTS OF IPC-2221, IPC-2222, AND IPC-6012 CLASS 2 5. ALL SPECIFICATIONS SHALL BE THE LATEST STANDARDS, UNLESS OTHERWISE NOTED 6. ALL MODIFICATIONS MUST BE COMMUNICATED AND APPROVED IN WRITING.				
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D	QA, ELECTRICAL TEST AND MARKINGS 15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
SILICONVALLEYGARAGE 				Project JWpulsegen-V3.PrjPCB Version: Variant [No Variations]	
				FABRICATION DRAWING	
1	2	3	4	5	6



COMPOSITE VIEW FRONT

A

A

B

B

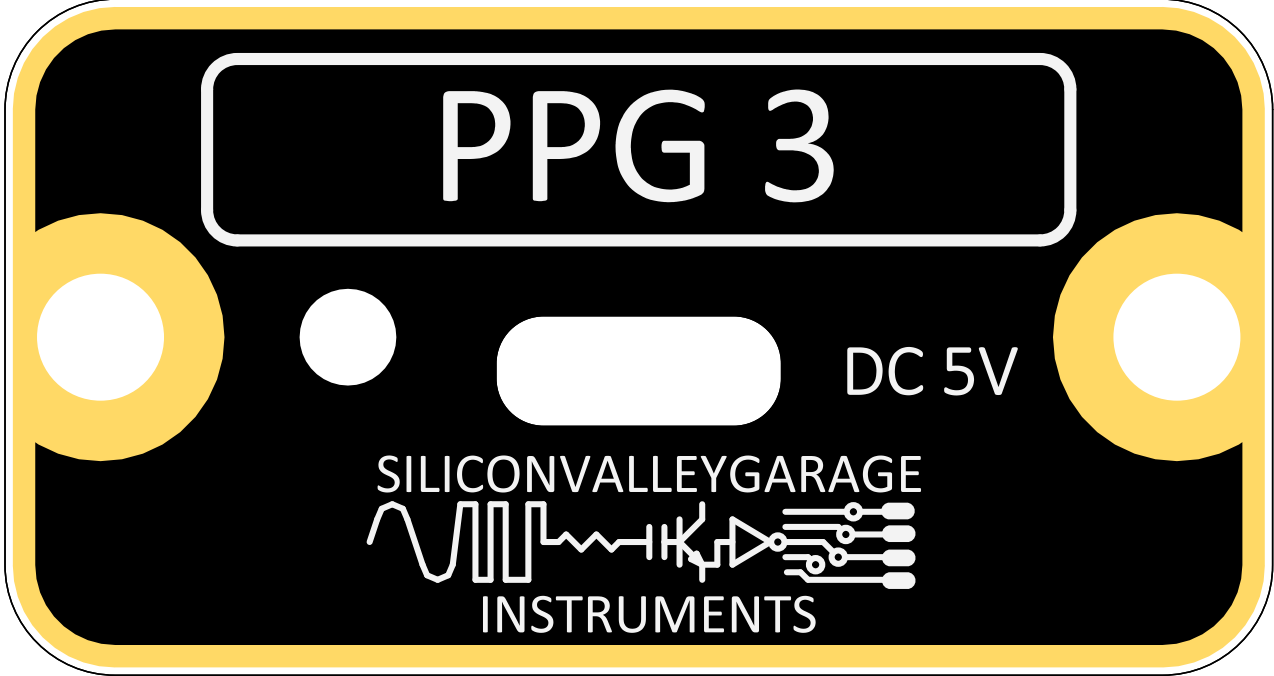
C

C

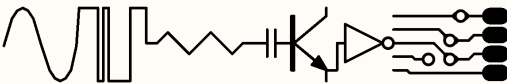
D

D

Top Layer (Scale 4:1)



SILICONVALLEYGARAGE



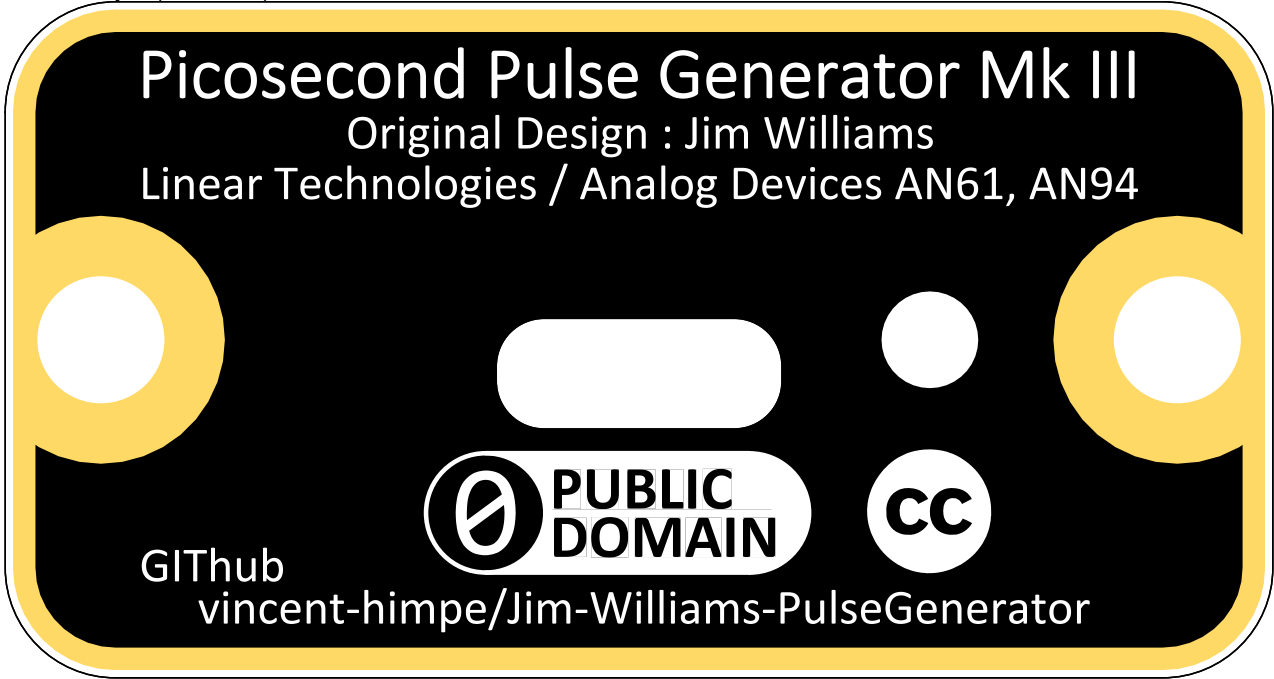
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

COMPOSITE VIEW BACK

Bottom Layer (Scale 4:1)



LAYER VIEW : TOP LAYER

A

A

B

B

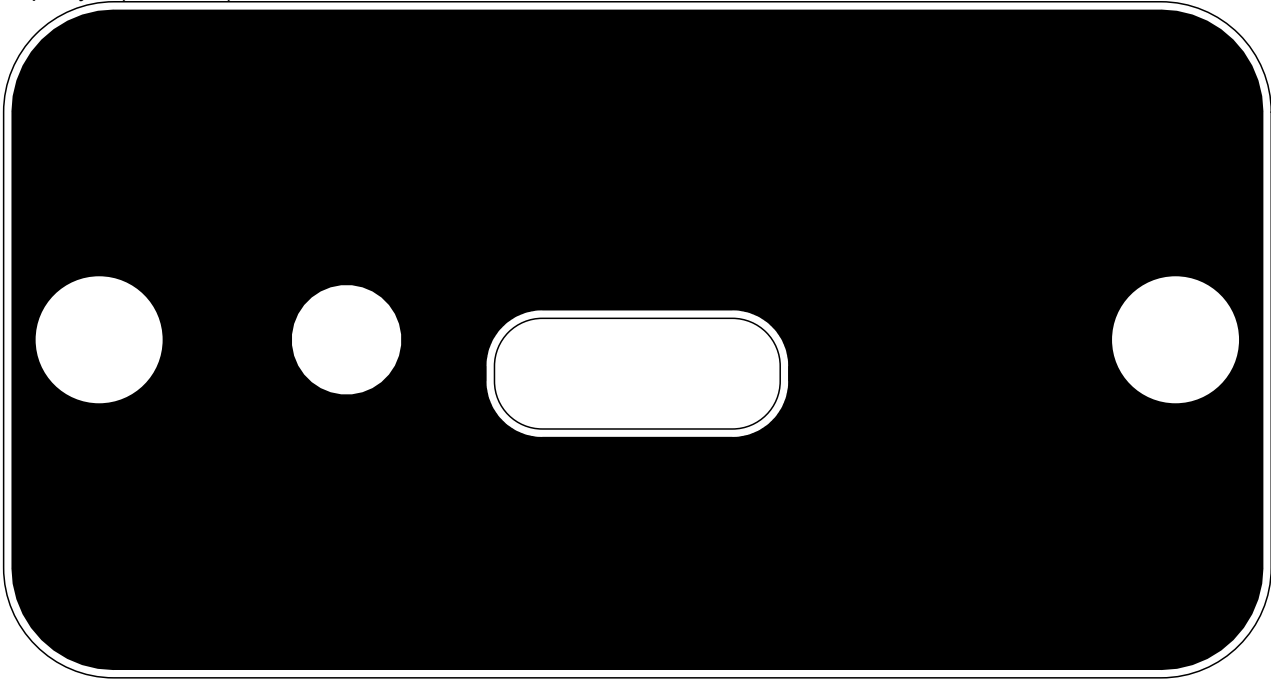
C

C

D

D

Top Layer (Scale 4:1)



LAYER VIEW : BOTTOM LAYER

A

A

B

B

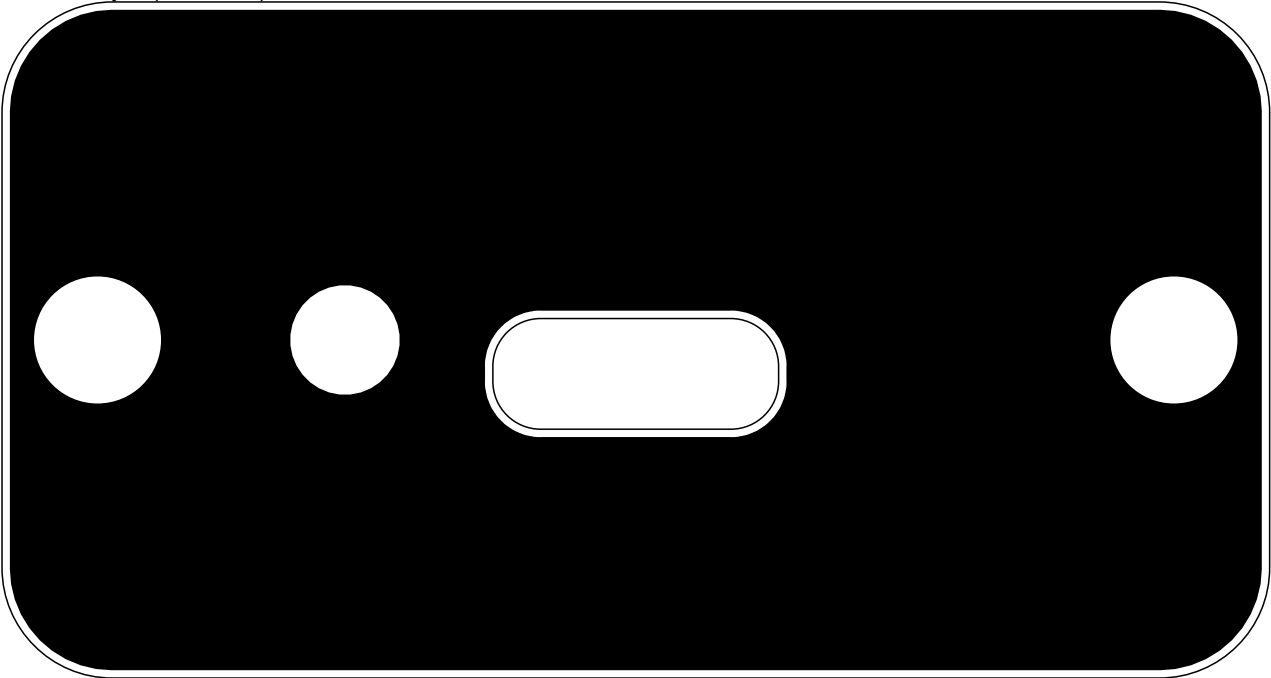
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C

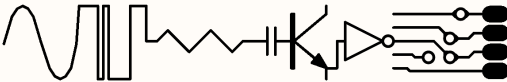
D

D

Bottom Layer (Scale 4:1)



SILICONVALLEYGARAGE



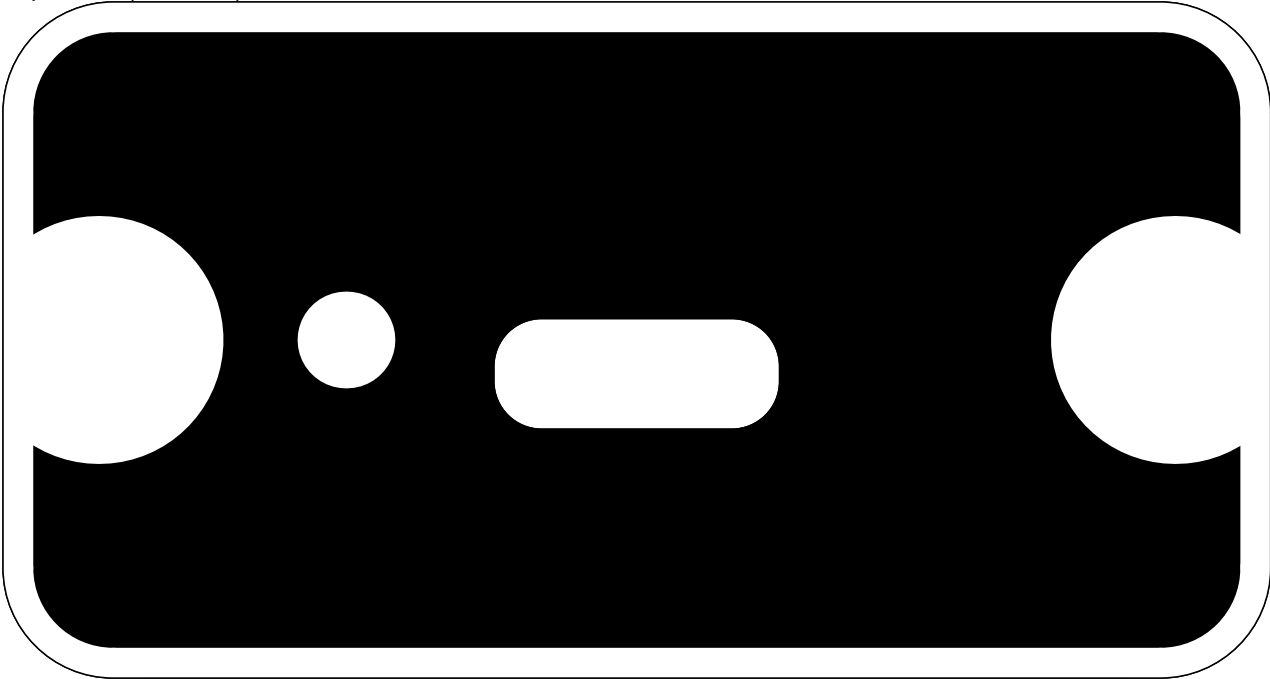
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

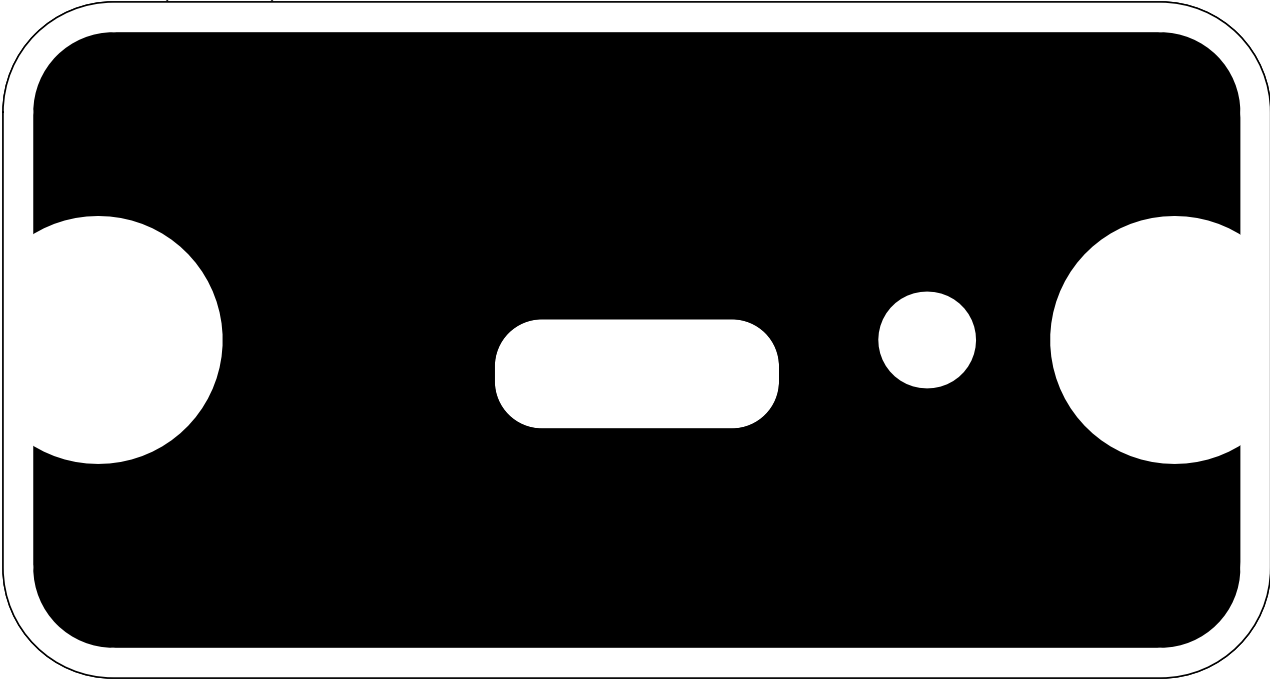
LAYER VIEW : TOP SOLDER MASK

Top Solder (Scale 4:1)



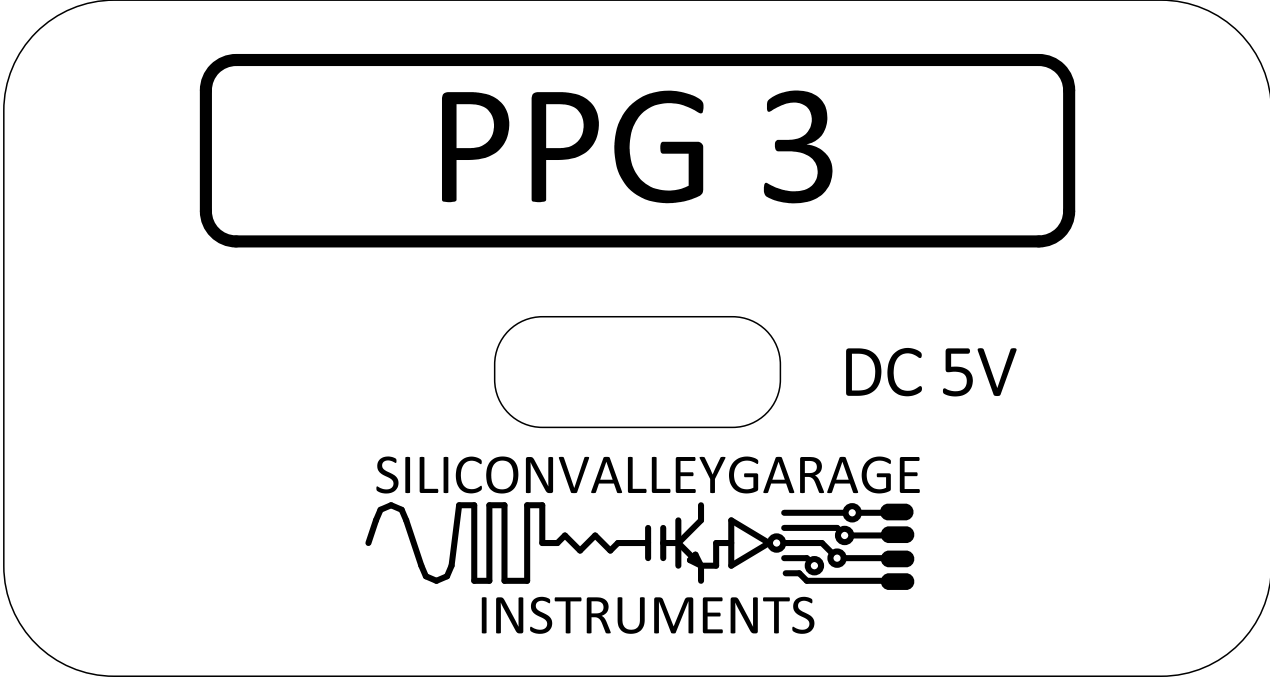
LAYER VIEW : BOTTOM SOLDER MASK

Bottom Solder (Scale 4:1)



LAYER VIEW : TOP SILKSCREEN (LEGEND)

Top Overlay (Scale 4:1)



LAYER VIEW : BOTTOM SILKSCREEN (LEGEND)

Bottom Overlay (Scale 4:1)

Picosecond Pulse Generator Mk III

Original Design : Jim Williams

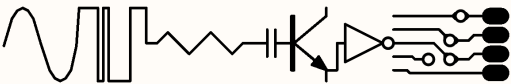
Linear Technologies / Analog Devices AN61, AN94

Public Domain

CC

GitHub

vincent-himpe/Jim-Williams-PulseGenerator

1	2	3	4	5	6
GENERAL					
A	GENERAL				
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C	11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm				
	12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL				
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	14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
	QA, ELECTRICAL TEST AND MARKINGS				
D	15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
SILICONVALLEYGARAGE				Project JWpulsegen-V3.PrjPCB	
				Version: Variant [No Variations]	
				FABRICATION DRAWING	
1	2	3	4	5	6

DRILL DRAWING

A

A

B

B

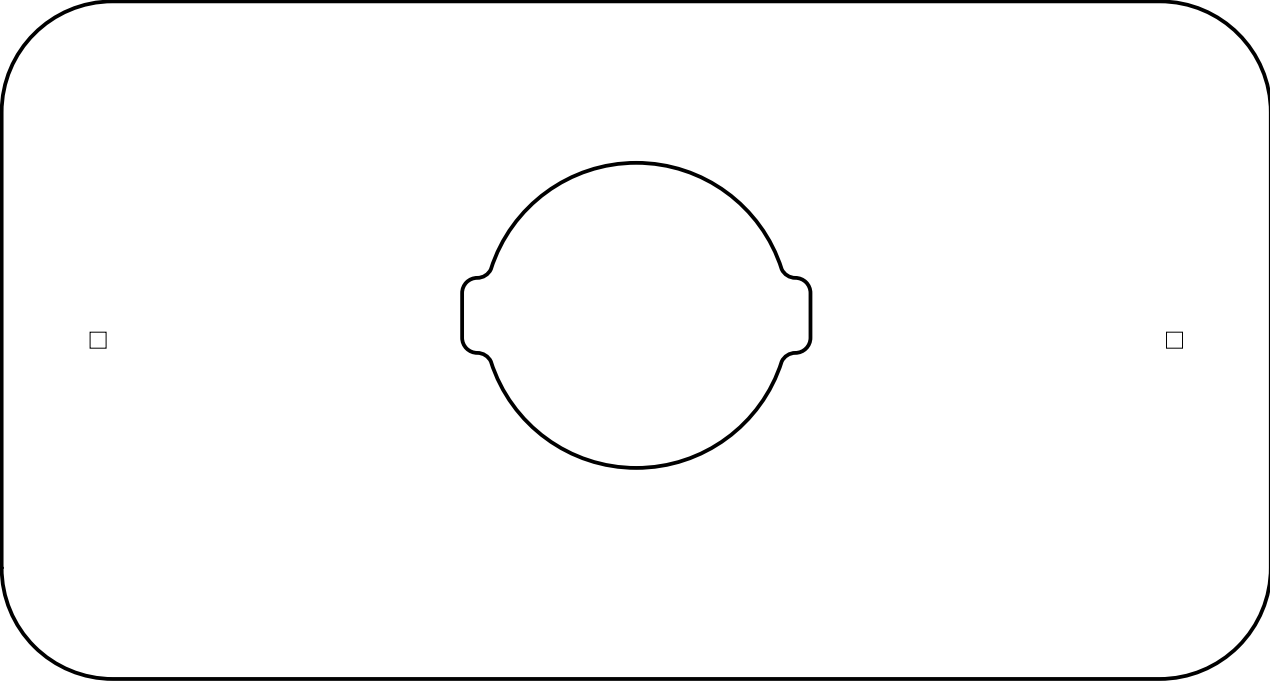
C

C

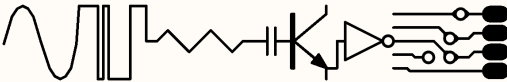
D

D

Drill Drawing View (Scale 4:1)



SILICONVALLEYGARAGE



Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

COMPOSITE VIEW FRONT

A

A

B

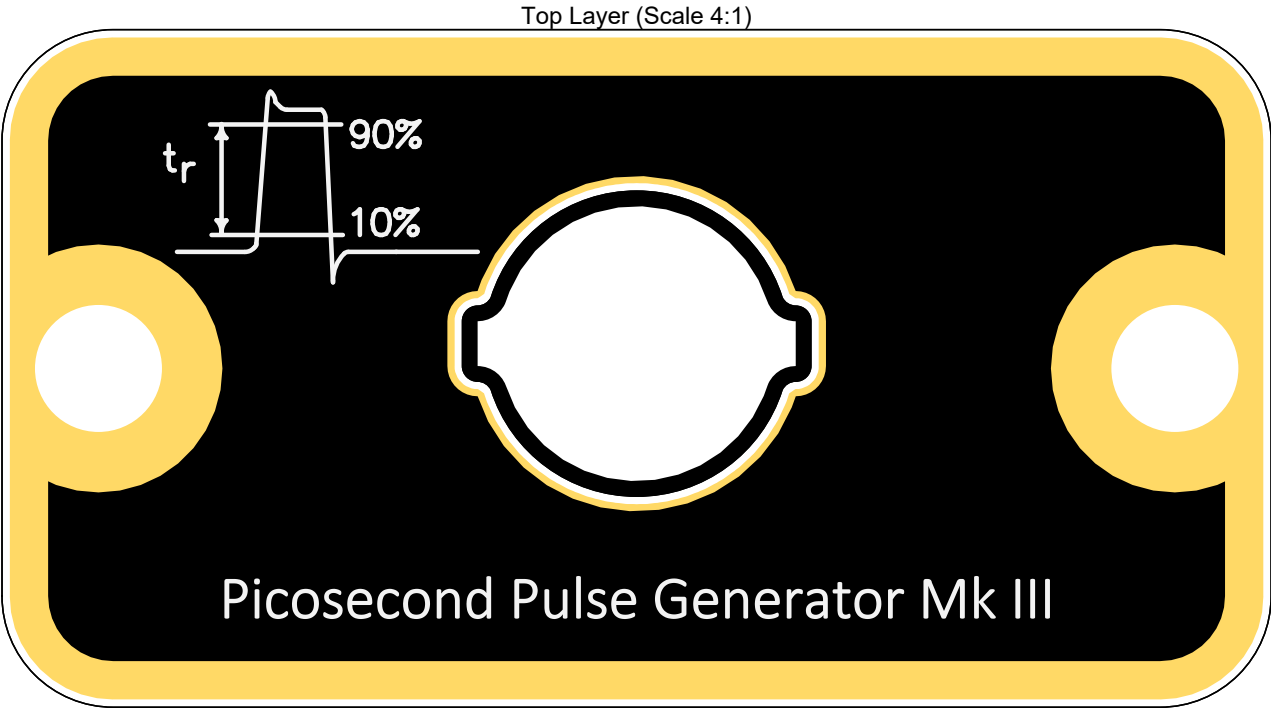
B

C

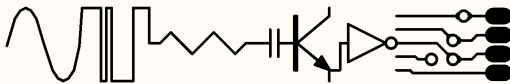
C

D

D



SILICONVALLEYGARAGE



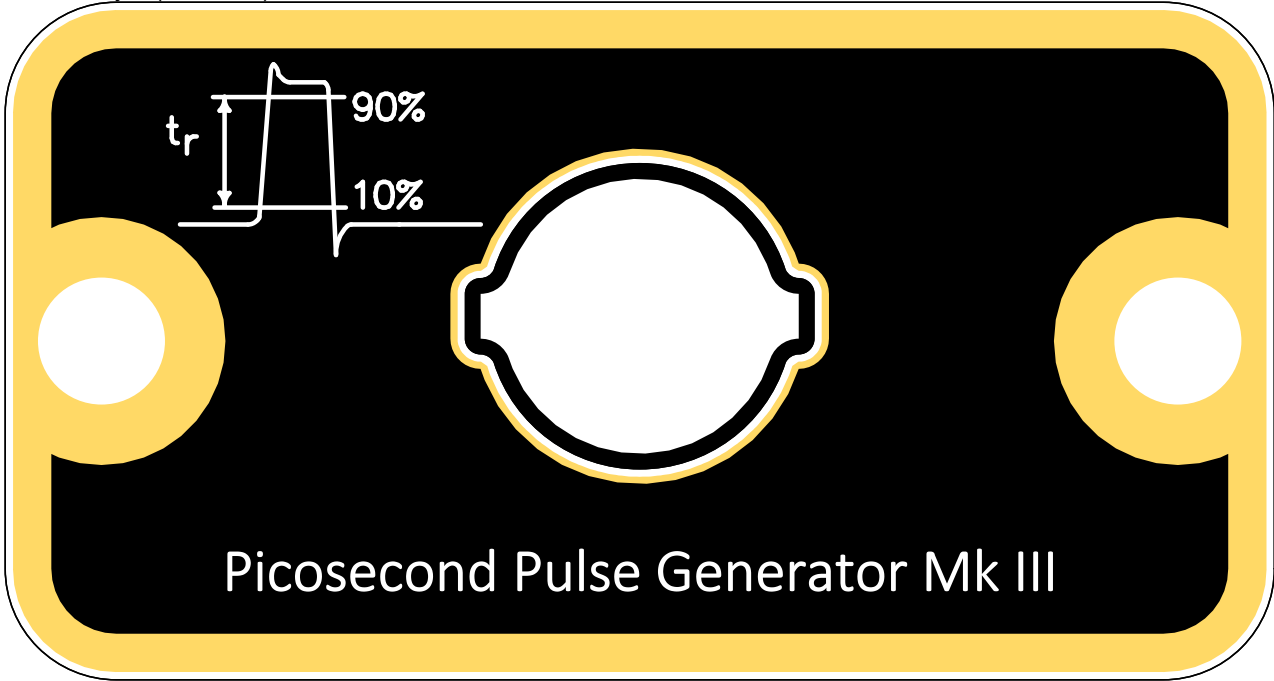
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

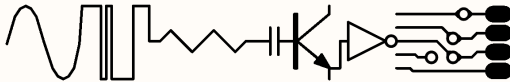
FABRICATION DRAWING

COMPOSITE VIEW BACK

Bottom Layer (Scale 4:1)



SILICONVALLEYGARAGE

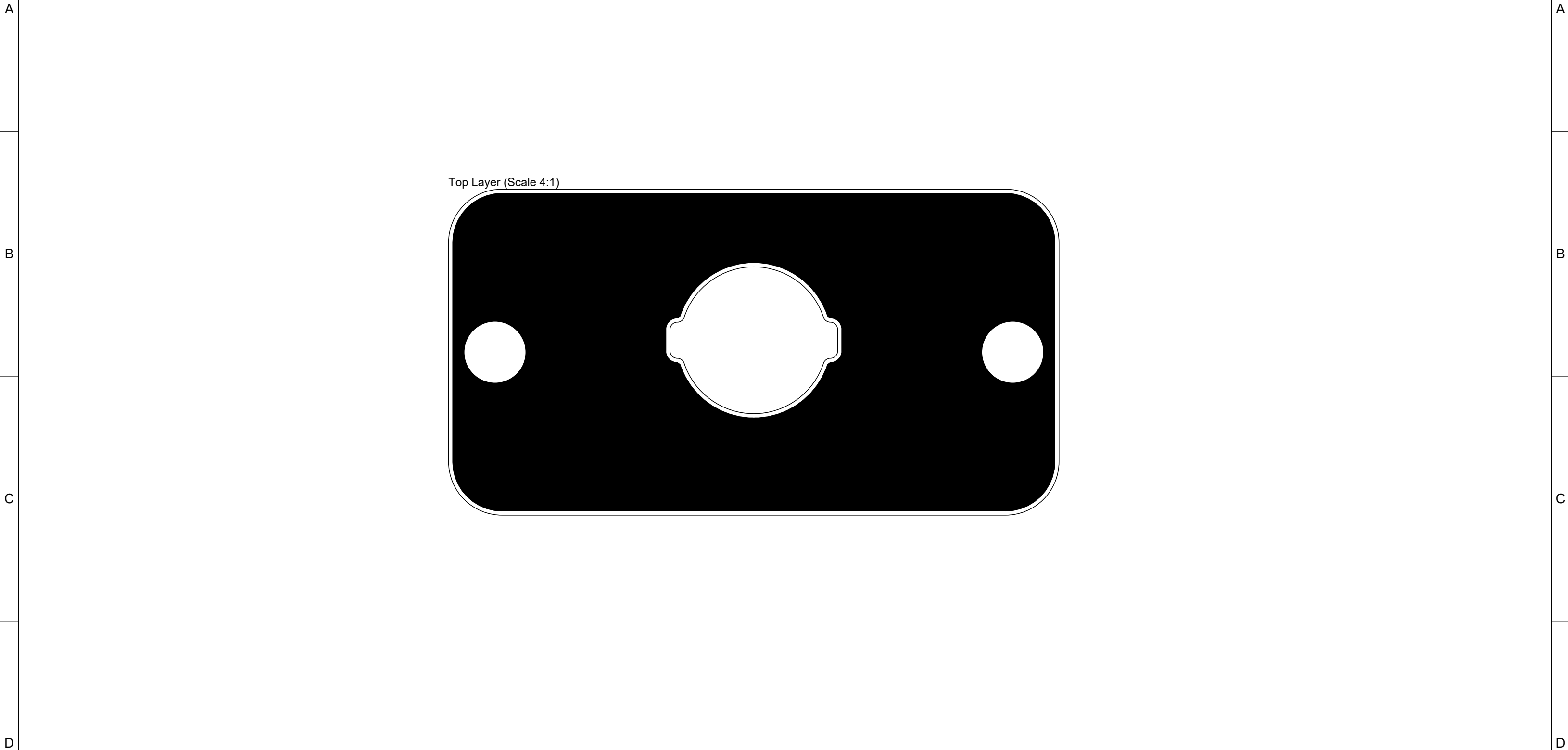


Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

LAYER VIEW : TOP LAYER



LAYER VIEW : BOTTOM LAYER

A

A

B

B

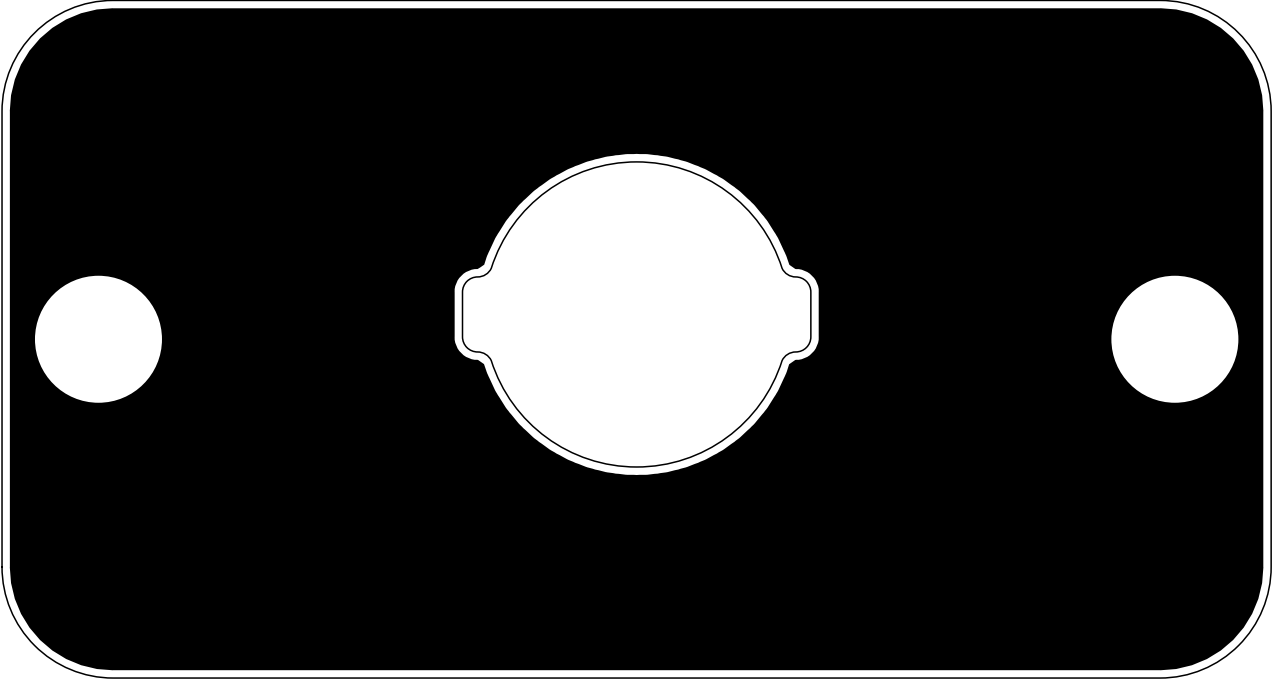
C

C

D

D

Bottom Layer (Scale 4:1)



SILICONVALLEYGARAGE

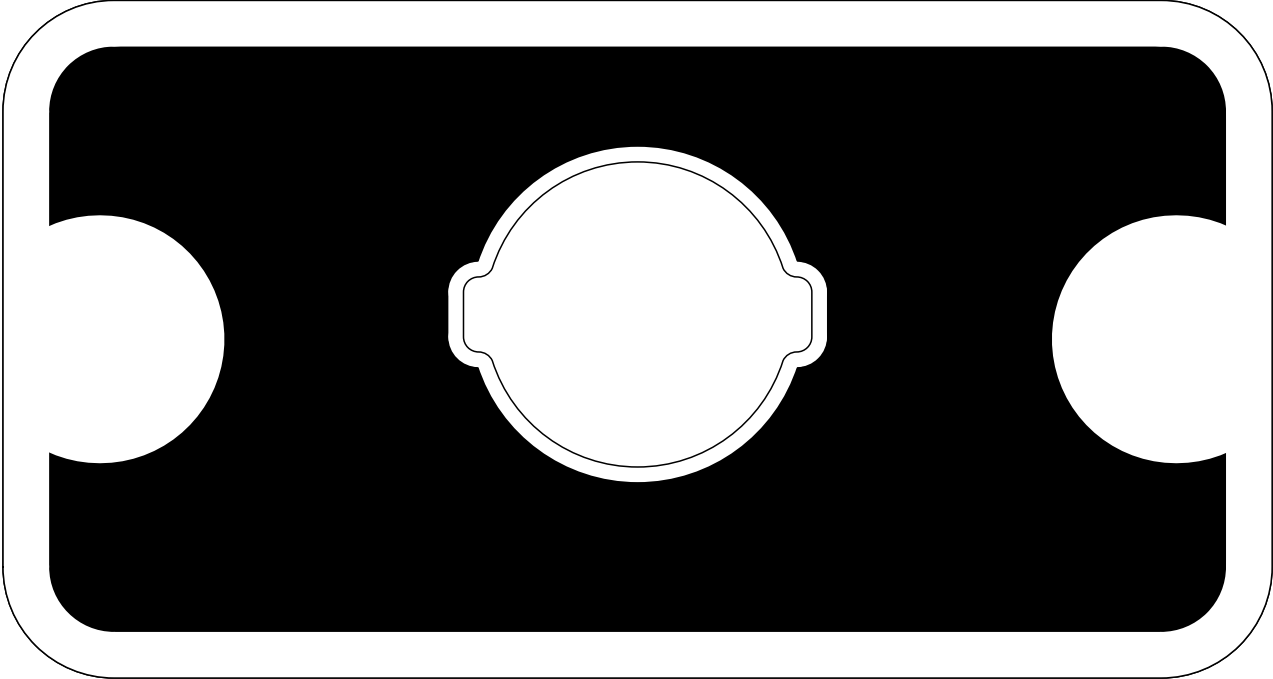
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

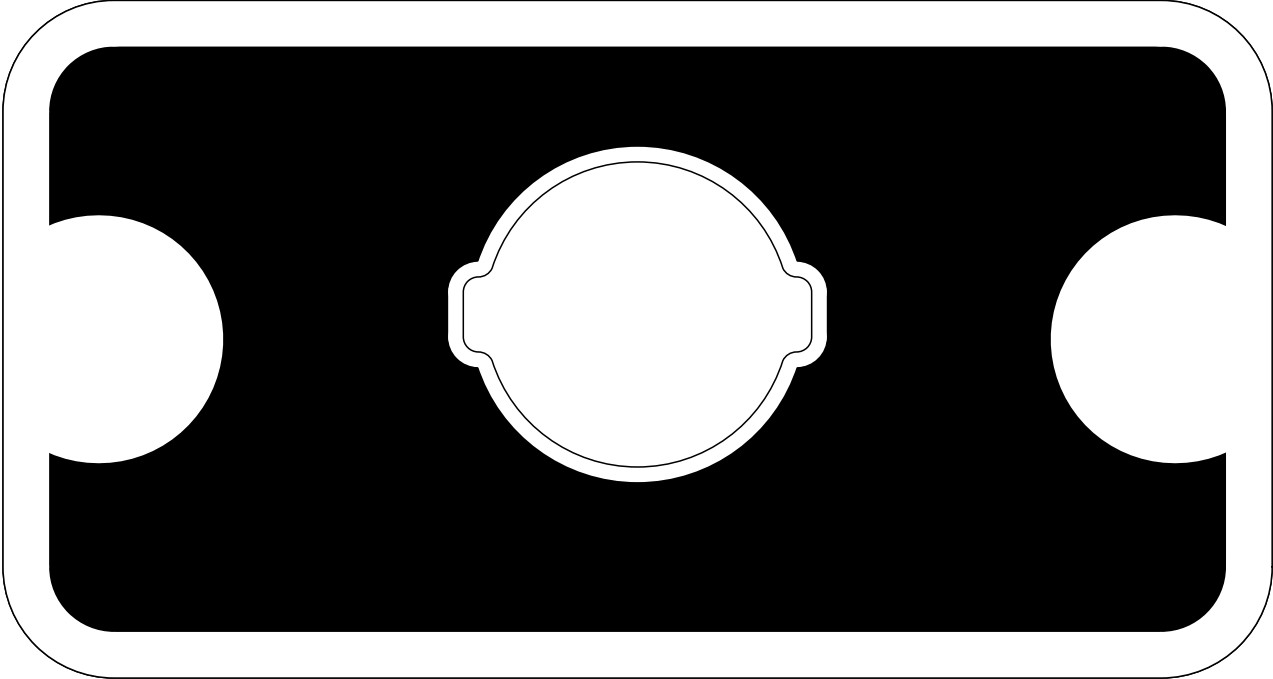
LAYER VIEW : TOP SOLDER MASK

Top Solder (Scale 4:1)



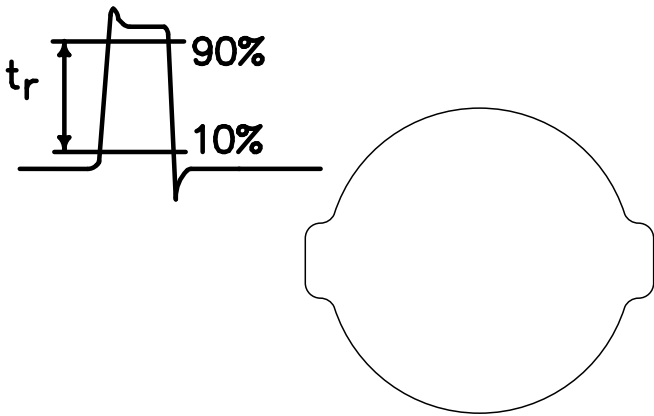
LAYER VIEW : BOTTOM SOLDER MASK

Bottom Solder (Scale 4:1)



LAYER VIEW : TOP SILKSCREEN (LEGEND)

Top Overlay (Scale 4:1)



Picosecond Pulse Generator Mk III

LAYER VIEW : BOTTOM SILKSCREEN (LEGEND)

